VAT Credit Refund, Corporate Liquidity and Financial Flexibility

Le Dai

School of Public Finance and Taxation,

Zhejiang University of Finance and Economics

Abstract

In 2018, China implemented the Value-added Tax (VAT) Credit Refund Policy, which was further expanded in 2022 to provide additional cash flow to businesses. This tax incentive policy significantly improved the financial condition of enterprises. This paper uses a composite indicator as a measure of corporate financial flexibility and treats the VAT refund policy as a quasi-natural experiment. Using data from A-share listed companies between 2015 and 2023, this study examines the impact of the VAT refund policy on corporate financial flexibility. The results indicate that the VAT refund policy significantly enhances corporate financial flexibility. Mechanism tests suggest that the policy improves financial flexibility by enhancing corporate liquidity, specifically by (1) improving cash flow management; (2) reducing current liabilities; and (3) increasing investment and research and development (R&D) expenditures to obtain long-term benefits, thereby enhancing financial flexibility. Dynamic effect analysis reveals that businesses first use the refunded VAT to repay short-term debt before expanding investments and R&D to enhance profitability. Finally, heterogeneity analysis shows that the policy has a more significant impact on the financial flexibility of firms with higher financing constraints, smaller sizes, and higher capital intensity.

Keywords: Value-added Tax Refund Policy; Financial Flexibility; Corporate Liquidity; Financing Constraints

1 Introduction

1.1 Policy Background

In the context of unprecedented global changes and the escalating trade tensions between China and the US, China's economy faces increasing downward pressure. The effective implementation of tax and fee reduction policies, combining both "counter-cyclical" and "pro-cyclical" fiscal policies, to activate market vitality while adhering to the principle of VAT neutrality has become a key topic in fiscal research. Enterprises are highly sensitive to external economic changes, and variations in external demand and the business environment are reflected in corporate financial and operational strategies. Especially during and after the COVID-19 pandemic, businesses have not recovered their profitability as quickly as expected by society. On the contrary, they have faced issues such as overcapacity, inventory accumulation, and reduced external demand. The accumulated VAT refund credits have exacerbated these problems, which, by occupying cash flow, indirectly reduce enterprises' risk resilience. In this context, enhancing corporate financial resilience, or financial flexibility, has become an urgent issue to address.

Value-added tax (VAT) is China's largest tax source, contributing nearly 40% of the national tax revenue following the "Business Tax to VAT"[1] reform in 2016. In 2023, domestic VAT revenue in China amounted to 6.93T RMB (962.5B USD), with an additional 1.84T RMB (255.6B USD) collected from VAT on imports and 1.71T RMB (237.5B USD) refunded through export VAT credit refunds. The total VAT revenue reached 7.06T RMB (980.6B USD), accounting for 39% of total tax revenue. In 2024, domestic VAT revenue reached 6.67T RMB (926.4B USD), representing 38% of total tax revenue. The VAT system in China operates on an invoice-based credit method, under which businesses serve as tax collectors and remit the net VAT to STA[2]. This process begins with the issuance of special VAT invoices for taxable transactions, which specify the output VAT charged to customers. These invoices enable buyers to claim input VAT credits. Businesses then calculate their VAT liability by subtracting the input VAT paid on purchases from the output VAT collected on sales. The resulting liability is determined by the formula below. A positive value indicates a tax payment due to the STA, while a negative value generates, a VAT credit arises.

$$\text{VAT}_{Payable} = \text{Output VAT} - \text{Input VAT} \tag{1}$$

Compared to other taxes, VAT is collected at the production stage, meaning it is levied on the cost side and has relatively strong cash flow characteristics. Prior to the introduction of the VAT Credit Refund policy, enterprises bore significant cash flow pressure. If businesses could timely obtain their refundable VAT credits, it would be crucial in improving their cash flow during periods of financial strain and debt pressure, promoting cash flow and cash holdings. In practical use, enterprises could repay debts, reduce current liabilities, and improve their credit ratings, thereby lowering financing costs. In times of emergencies, these funds can be considered as part of the company's own capital, helping the company mitigate losses, especially for industries and enterprises that have high input VAT but low sales (i.e., low output VAT), such as agriculture and machinery manufacturing (T. Zhang and Liu (2024)). The financial situation of start-up enterprises, which require significant investment but have not yet achieved economies of scale, has also improved significantly. This aligns with the government's policy direction of supporting small and micro enterprises and high-tech small and micro enterprises. Therefore, VAT credit refund policy was introduced and it aims at resolving these challenges. As a result, the question of whether and how the VAT refund policy can enhance financial flexibility has become a topic worthy of research.

The VAT credit refund policy stemmed from the "structural tax reduction" concept proposed by the central government, targeting specific taxes to reduce tax burdens and promote economic transformation. As China's economy gradually enters a normalization phase, the country faces social contradictions such as increasing demand that mismatches the supply quality. The industrial structure, which was once driven by labor-intensive sectors, is beginning to decline, with the labor dividend gradually disappearing and the fertility rate decreasing. These various issues have prompted a shift towards supply-side structural reforms. These reforms aim to reduce the tax burden on enterprises and provide incentives for innovation, expansion, and environmental protection investment, thus lowering costs. To address the problem of double taxation, the government initiated the "business tax to VAT" policy in 2012 and fully implemented it in 2016. This policy eliminates the double taxation of business taxes and allows VAT to be deducted

[1]The Transition from Business Tax to VAT refers to a significant change in a country's tax system, where the taxation on business activities is shifted from a business tax model to a VAT system. Under the business tax model, taxes are levied on the total value of the transaction. However, in China's VAT system, taxes are applied incrementally at each stage of production or distribution based on the value added to goods or services. This shift aims to reduce economic distortions, improve transparency, and broaden the tax base. See https://fgk.chinatax.gov.cn/zcfgk/c102440/c5207018/content.html.

[2]State Taxation Administration, P.R.China. See https://www.chinatax.gov.cn/eng/home.html.

throughout the entire industrial chain, improving the transparency and efficiency of the tax system.

The VAT Credit Refund policy, also known as the VAT refund for excess input tax policy, allows businesses to reclaim the excess value-added tax (VAT) paid on inputs over the VAT collected on their outputs. This refund is an essential tool for enhancing business liquidity, especially for businesses involved in capital-intensive industries, such as manufacturing and high-tech sectors. The policy is designed to support business operations, promote investment, and stimulate economic growth by reducing the tax burden on companies that have paid more VAT on purchases than they have collected on sales. This refund mechanism has been a vital part of China's tax policy reforms, particularly for businesses that have a significant amount of input tax due to high capital expenditure or export activities. It encourages economic activity by enabling companies to improve cash flow, which can be reinvested in the business.

The VAT Credit Refund Policy constitutes a key component of China's tax reduction and fee-cutting framework. It targets the value-added tax system by refunding the portion of unutilized input VAT credits that remain at the end of the accounting period, thereby directly enhancing firms' cash flow. In practice, when a firm's input VAT exceeds its output VAT liability, the remaining unused credits are refunded without charge, alleviating liquidity constraints.

From a policy evolution perspective, VAT credit refunds were first explicitly introduced in June 2018 under the Ministry of Finance and State Administration of Taxation Document No. 70 [2018], initially limited to 18 designated industries. Since 2020, in response to the economic shocks of the COVID-19 pandemic, coverage has gradually expanded. Beginning in 2022, the policy applied to 16 industries, including advanced manufacturing as well as micro and small enterprises, and was extended further to wholesale, retail, and other sectors.

From a historical perspective, the VAT refund policy was first introduced in the circular "MOF[3]&STA [2018] No.70"[4] issued in June 2018, initially targeting 18 specific industries. In 2020, in response to the impact of the pandemic on business operations, the scope of the policy was expanded to cover 16 industries, including advanced manufacturing, micro, small enterprises, as well as wholesale and retail sectors. The scale of VAT refunds has shown an increasing trend year by year. According to the State Administration of Taxation, from 2019 to 2021, the total VAT refund amounted to 1.2T CNY (167B USD). In 2022, the policy expanded till another 16 industries according to MOF&STA [2022] No.14 [5] and the total refund reached 2.3T CNY (320B USD), with the manufacturing industry, especially small and micro enterprises, being the major beneficiaries. This policy has reduced the financing pressure faced by small and medium-sized enterprises, especially evident during the economic recession. The evolution of the VAT refund policy reflects China's tax system's transition from a production-based VAT system to a consumption-based VAT system and highlights the flexible application of tax policies in macroeconomic regulation. The continuous improvement of the policy not only optimizes the business environment but also provides strong support for high-quality economic development. In the future, with the further digitization and automation of tax administration systems, the VAT refund policy is expected to be implemented more efficiently. The history of VAT credit refund policy is summarized as the Table 1.1 shown below. For detailed policy document, please refer to the translated official documents. For year 2018, see Appendix E; For year 2022, see Appendix F attached in the end.

[3]Ministry of Finance, P.R.China. See https://www.mof.gov.cn/en/.
[4]See https://www.mof.gov.cn/gkml/caizhengwengao/wg2018/wg201807/201810/t20181023_3052987.htm.
[5]See https://www.gov.cn/zhengce/zhengceku/2022-03/22/content_5680600.htm.

Table 1.1: Implement History of VAT Credit Refund Policy

No.	Official Document No.	Title	Key Content
1	MOF&STA[2018]No.70	Notice on Policies for VAT Credit Refunds in Certain Industries in 2018	Priority refund to 18 industries, including advanced manufacturing and modern service sectors; refund retained input VAT.
2	MOF&STA&GAC[2019] No.39	Announcement on Policies Related to Deepening VAT Reform	Further improvement in VAT credit refund mechanisms; full refund for eligible enterprises.
3	MOF&STA[2019]No.84	Announcement Clarifying Refund Policies for Advanced Manufacturing VAT Credits	Clarifies criteria for advanced manufacturing industries to enjoy VAT refunds.
4	MOF&STA[2020]No.4	Announcement on Tax Policies Supporting Epidemic Response	Full refund of incremental VAT credits for affected enterprises in response to COVID-19.
5	MOF&STA[2021]No.15	Announcement on VAT Credit Refunds for Advanced Manufacturing in 2021	Further expansion of scope and refund amount for advanced manufacturing.
6	MOF&STA[2022] No.14	Announcement on Enhancing VAT Credit Refund Implementation	Expands scope to large, medium, and small enterprises; includes stock and incremental refunds.
7	MOF&STA[2022]No.21	Announcement on Expanding to 16 Industries Eligible for VAT Credit Refunds	Further expansion to more industries, including wholesale and retail sectors.

Among major developed economies worldwide, similar VAT credit refund mechanisms exist to address the imbalance that arises when firms' input VAT exceeds their output VAT during production and business operations. Within the European Union's VAT framework, the EU VAT Directive requires member states to provide refund channels, allowing firms under certain conditions to reclaim excess input VAT. In countries such as Germany and France, firms are required to submit refund applications at the end of each tax period. Keen and Lockwood (2010) demonstrate that refund mechanisms safeguard the neutrality principle of VAT and enhance corporate cash flow. De la Feria (2018), in a comparative study, highlights the correlation between fiscal and administrative capacity and refund efficiency, further arguing that refund policies facilitate international trade. Many advanced economies adopt relatively lenient conditions for SMEs in accessing VAT refunds and leverage digitalized systems to improve efficiency. Such measures alleviate financing constraints for SMEs, an effect that becomes particularly salient during periods of economic downturn Bird and Gendron (2007). Nevertheless, some literature has cautioned that overly relaxed refund regimes may create vulnerabilities to tax fraud (Tait (1988)).

1.2 Concepts and Definitions

I. Definition of Financial Flexibility

Enterprises are the cells of the national economy, and the financial status of individual enterprises reflects the overall business environment, the enthusiasm of policies, and the pervasiveness of an innovation culture. The literature focusing on corporate financial conditions mainly considers the following indicators: net profit, operating cash flow, and financing constraints. However, these single-item indicators can only reflect one aspect of a company's financial status. For example, operating cash flow can only reflect the inflow and outflow of cash, but it does not lead to a conclusion that the company's financial condition

is favorable. High cash flow may arise from asset sales or external financing, and an increase in financing could lead to debt accumulation, worsening financial conditions. Companies may also accumulate a significant amount of profit through promotional activities in a short time, but long-term profitability may not be ideal. The effective utilization of funds cannot easily be reflected by net cash flow either. Therefore, the financial condition of a company needs to be measured using a composite indicator, which combines both long-term and short-term dimensions and accounts for cash flow and debt.

The financial flexibility explored in this paper primarily refers to the ability of enterprises to respond to external shocks, used to evaluate their overall financial health. The pioneering research of Gamba and Triantis (2008) provided a foundational framework for defining financial flexibility, stating that financial flexibility is exhibited in the ability of enterprises to adjust their capital structure, manage cash reserves, and utilize financing channels. Specifically, it is reflected through cash holdings, external financing costs, and debt repayment capacity. Over time, research on financial flexibility has shifted from purely theoretical analysis to empirical studies based on company data. However, there is limited literature examining changes in financial flexibility under pure tax policy shocks, their transmission mechanisms, and their heterogeneity effects, especially regarding Chinese enterprises in the context of major global changes. As China and the global economy face increasing downward pressure, escalating conflicts, and the intensification of the Sino-US trade war, enterprises have encountered external pressures such as reduced orders, investment restrictions, and insufficient consumer spending, while internally facing poor management, inventory accumulation, and financing obstacles. The study of corporate financial flexibility, particularly the ability of firms to resist external risks, has received growing attention as a core component of long-term corporate strategy DeAngelo and DeAngelo (2007). Researching how enterprises manage financial flexibility under increasing risks—both internal and external—has important practical significance. This includes (1) guiding enterprises in developing investment and financing plans, and how to effectively use tax incentives to adjust production and sales plans; (2) helping the government gain a more comprehensive and systematic understanding of enterprise financial risks (beyond just financial statements) and improving policy coverage, scope, and heterogeneity for different types of enterprises, thus enhancing policy implementation efficiency and economic resilience.

II. Definition of Corporate Liquidity

Corporate liquidity is a popular topic in corporate finance and financial management, primarily used to measure a company's ability to repay debts in the short term. This indicator provides meaningful information to assess a company's financial health, offering reference points for company management. The key aspect of liquidity lies in the ability of a company's assets to be converted into cash and the company's ability to meet short-term debt obligations. According to the definition provided by Ross et al. (2019), liquidity is closely related to a company's short-term assets and liabilities, emphasizing the time-related and controllable nature of liquidity. That is, liquidity focuses on whether a company has the ability to raise funds promptly when it faces payment demands. Compared to financial flexibility, which concerns both short-term and long-term debt repayment abilities, liquidity is more focused on short-term solvency. The financial flexibility and liquidity discussed in this paper are interrelated, with liquidity being one aspect of financial flexibility. This paper will use empirical research to demonstrate how the VAT Credit Refund policy can improve corporate financial flexibility by enhancing corporate liquidity.

The VAT Credit Refund policy theoretically promotes financial flexibility, whether from the perspectives of cash flow, debt management, or profitability. The theoretical conclusion is consistent across these aspects. Studying the impact of the VAT Credit Refund policy on corporate financial flexibility, combining both short-term and long-term effects, conducting policy heterogeneity analysis, and analyzing the transmission mechanism of this policy has strong practical significance.

1.3 Structure

This paper examines the relationship between the VAT Credit Refund policy and corporate financial flexibility. Section 1 introduces the policy background and the conceptual foundation of financial flexibility. Section 2 reviews the relevant literature.Section 3 proposes the theoretical hypotheses. Sections 4, 5, and 6 present the empirical analysis, including baseline regression, robustness checks, and further analyses such as mechanism and heterogeneity testing. The seventh section concludes the paper and offers policy recommendations.

2 Literature Review

Based on the theme of this study, this section organizes the literature review into several modules for summary and synthesis, beginning with definitions and measurements of firm financial flexibility, followed by a review of general tax incentive policies and their relationship with financial flexibility. Subsequently, the literature regarding the VAT Credit Refund policy will be summarized. Finally, we examine the concept of corporate liquidity. These components are integrated in the final part to provide a comprehensive assessment of the existing literature, ensuring full coverage of the research context.

2.1 Literature on Tax Incentive Policies

In the macro perspective, VAT Credit Refund policy falls under the category of tax incentive policies, as it effectively reduces the tax burden on enterprises. Tax incentive policies typically aim to encourage firms and individuals to engage in specific behaviors—such as increasing investment, promoting technological innovation, or attracting foreign capital—through tax exemptions, credits, or refunds. This concept is rooted in classical economic theories, including investment theory Jorgenson (1963) and tax competition theory Wilson (1999), which emphasize the strategic use of tax rates by countries and regions to attract capital and enhance competitiveness. These policies play a vital role in shaping economic structure, promoting productivity, and improving corporate financial conditions.

"Tax Incentives for Business Investment: A Primer for Policy Makers in Developing Countries" (2002) point out that tax incentives can reduce the tax burden on specific stages of business activity, and come in various forms, including tax holidays, credits, and rate reductions. Such policies are widely used across countries. For instance, the United States implements state-level corporate tax credits and provides targeted incentives for specific industries Cailin and Zidar (2020).

From a macroeconomic perspective, tax incentives produce effects on multiple levels. At the firm level, they stimulate investment expansion. At the macro level, they can increase foreign direct investment (FDI) and encourage innovation (World Bank (2020)). However, the actual effects of tax incentives do not always align with expectations. While most studies find positive effects on growth, others report limited or even negligible impacts on firm investment, financial performance, or broader economic outcomes. A World Bank (2020) report found that nearly half of developing countries implemented tax incentive policies. Gravelle (2006) and Gravelle (2014) analyzed their impacts in OECD countries, especially the U.S., noting that such policies increased national employment from 66 million in 1961 to 117 million in 1991. Their quantitative model shows that reducing the corporate tax rate from 35% to 25% could modestly raise wages and output, increasing firm output by 0.2%.

However, Yagan (2015) found that the 2003 U.S. capital tax reform did not significantly stimulate corporate investment or income growth as expected. Similarly, Mutti and Grubert (2004) showed that tax incentives did not encourage multinationals to invest in developing countries.

In recent years, studies have focused on the U.S. Tax Cuts and Jobs Act (TCJA) of 2017, which reshaped the U.S. tax system by (1) lowering the corporate income tax rate from 35% to 21%, (2) adjusting personal income tax rates, and (3) revising deductions and credits (Gale et al. (2018)). The act raised standard deductions and limited certain individual income tax deductions. From a macro perspective, the policy led to a short-lived economic "boom," but its long-term effects remain uncertain (Zidar et al. (2018); Desai et al. (2020)). Furthermore, scholars have warned about the potential fiscal costs of expanding tax incentives. Klemm (2021) emphasized administrative and compliance costs and the risks of firms exploiting tax breaks through inflated reporting, leading to resource waste and unfair competition. The World Bank (2020) also warned that such policies might cause fiscal crises in low-income countries that fail to attract enough investment or have weak administrative capacity.

At the micro level, the benefits of tax incentives are most visible in corporate operations. Essentially, tax incentives lower effective tax rates, enabling firms to allocate more cash toward operations, expansion, or debt repayment. As part of proactive fiscal policy and cost reduction, tax cuts are a critical component of China's current and future economic strategy. These policies aim to (1) act as counter-cyclical tools to stabilize output and (2) support supply-side structural reforms by stimulating corporate vitality and improving total factor productivity (Zwick and Mahon (2017); Y. Liu and Mao (2019); L. Tian and Lu (2021)).

Research also finds that the effects of tax incentives are heterogeneous—depending on firm size, industry, financing constraints, and firm characteristics. Yet, findings remain mixed. Zwick and Mahon (2017) grouped firms by size and found that small firms increased investment more significantly after receiving tax incentives. Desai and Dharmapala (2009) argue that large multinationals benefit more from tax planning, while small and medium-sized enterprises (SMEs) lack such capacity. Similarly, Hassett and Hubbard (2002) found that tax incentives have a greater impact on total factor productivity (TFP) in smaller firms. Regarding financing constraints, Rauh (2006) showed that firms facing severe constraints tend to use tax savings for liquidity rather than long-term investment—while the opposite holds for less constrained firms.

2.2 VAT Refund and Corporate Liquidity

Corporate liquidity reflects a company's ability to meet its short-term financial obligations. This ability is usually measured by the relative relationship between the company's cash flow and the short-term liabilities it bears. Existing literature suggests that companies with abundant cash flow and lower short-term debt pressure tend to exhibit better liquidity, which allows them to better respond to external shocks or investment opportunities (Acharya et al. (2007); Almeida et al. (2004); Faulkender and Wang (2006)). Acharya et al. (2007) note that cash flow can serve as a hedging tool against the risks posed by liabilities, meaning that firms with sufficient cash flow are better able to withstand liquidity shocks caused by short-term liabilities, thereby showing higher liquidity levels. Almeida et al. (2004) examined the relationship between cash flow sensitivity and cash holdings, finding that companies with ample cash flow are better positioned to accumulate cash reserves to cope with potential future financing constraints or debt repayment pressures, thereby demonstrating superior liquidity. Faulkender and Wang (2006) further emphasized that changes in cash flow have a significant impact on firm value, as cash flow directly reflects a company's ability to meet short-term debt obligations. Companies with strong cash flow and a lower proportion of short-term debt typically possess stronger debt repayment capacity, which enhances their valuation in the capital market and reflects investor recognition of the company's liquidity.

2.3 Literature Review on the VAT Credit Refund Policy

Most studies concerning VAT Credit Refund Policy centered on the Document No.70 issued by the MOF and STA in 2018, and use this document as the intervention point for empirical analysis. This document explicitly defined 18 pilot industries for the policy trial, allowing for a clear division between treatment and control groups. Based on this classification, researchers often construct difference-in-differences (DID) models to better estimate policy effects and assess changes in firms' operational performance.

The existing research on the effects of the VAT credit refund policy can be categorized into two main levels: the firm level and the macroeconomic level. At the firm level, the policy is found to influence financial performance, total factor productivity (TFP), and investment behavior. At the macro level, attention has primarily focused on employment outcomes. At the firm level, the policy is shown to improve profitability and firm value—often the dependent variables in empirical models—by easing financial constraints and encouraging increased investment. Some studies even find the policy helps mitigate financial distress. For example, a research team from the State Taxation Administration of Gansu Province (Gansu Provincial Tax Bureau Research Group et al. (2024)) found, using industrial firm data from Gansu, that the VAT credit refund significantly stimulated corporate investment. Most studies, however, treat investment as an intermediate mechanism rather than a final outcome. Wu et al. (2021) analyzed firm-level stock returns around the policy announcement window and demonstrated that the increase in firm value could be explained by an expansion in investment, a finding supported by He et al. (2019).

Several recent studies have also examined how the policy encourages green innovation and ESG-related R&D investments. Wang Pei et al. (2024) and Fang Xiaobin (2024) found that VAT refunds significantly boosted environmentally friendly innovation. T. Zhang and Liu (2024) revealed that the policy effectively alleviated firms' financing constraints and promoted upstream-downstream industrial linkages. Interestingly, H. Cui et al. (2022) reported a "crowding-out" effect, noting that increased refunds may not always translate into proportional increases in investment. Some studies have directly assessed the policy's effect on firms' financial health. Xie et al. (2022) found that VAT credit refunds reduced cash holdings, with more pronounced effects among firms with extremely low or high cash reserves—those at the tails of the distribution. At the macroeconomic level, the policy's effect on employment has yielded mixed findings. Z. Liu et al. (2023) argued that the refund policy might crowd out employment, while others, like X. Cui et al. (2023), found the opposite—firms expanded employment in response to increased liquidity, suggesting a positive employment effect.

The effectiveness of the VAT refund policy is influenced by firm size, ownership structure, industry characteristics, regional policy environments, and environmental regulations, creating substantial heterogeneity in outcomes. Larger firms appear to benefit more from the policy due to their greater capital needs and capacity to use refunded VAT to ease financial constraints (Yue and Xiao (2023); T. Zhang and Liu (2024)). Industry-wise, capital- and technology-intensive sectors benefit more, especially in R&D investment, while labor-intensive sectors and firms with fewer financing constraints benefit less. Cui Huiyu et al. (2022) argued that private firms, tech-intensive industries, and large enterprises experienced stronger crowding-out effects in their R&D investment. Zhao et al. (2024) noted that labor-intensive firms had lower compliance in social security contributions, adding another layer to the interpretation of financial health. Moreover, existing literature finds that firms with lower cash flows and working capital turnover ratios tend to benefit more, as they are more cash-constrained and more reliant on VAT refunds for liquidity relief (Y. Zhang et al. (2024); Wang and Li (2024)). The effectiveness of the policy also varies with regional tax structures, fiscal burdens, and the administrative capacity of local governments. Z. Tian et

al. (2024) highlighted inter-regional competition in tax refund implementation, which may distort policy outcomes.

2.4 Literature Review on Corporate Liquidity

Corporate liquidity originates from several foundational economic theories. First, Keynes' theory of liquidity preference (Keynes, 1936)Keynes (1936) outlines multiple motives for holding liquid assets, including speculative and precautionary needs. Second, the cash management model developed by M. H. Miller and Orr (1966) provides a deeper analysis of liquidity optimization by assuming stochastic cash flows and proposing upper and lower bounds for cash holdings, offering a framework for balancing liquidity with opportunity cost. Third, trade credit theory (Ferris (1981)) suggests that firms can strategically delay payments to suppliers or accelerate collections from customers to enhance short-term liquidity. It can be measured through a variety of indicators. From a capital structure perspective, metrics such as the current ratio or quick ratio are commonly used. From a cash perspective, liquidity is often proxied by operating cash flow. Some studies also incorporate stock market liquidity, particularly in the context of publicly listed firms Amihud (2002). According to liquidity theory, an increase in liquidity tends to reduce financial and operational risk, while simultaneously encouraging investment expansion Myers and Majluf (1984). However, other scholars caution that excessive liquidity may lead to inefficient capital allocation and resource hoarding (Opler et al. (1999); Eljelly (2004)).

Corporate liquidity can be influenced by a wide range of internal and external factors. Internally, firms with lower liquidity often exhibit slower growth due to financial constraints that hinder asset expansion and revenue growth. However, small firms may leverage short-term debt expansion to rapidly scale up in the short termC. Liu et al. (2023). Ni et al. (2024) emphasized that if nominal liquidity lacks purchasing power, firms may resort to increased short-term financing. Luo and Liu (2021) argued that financial asset allocation can both provide liquidity to ease financing constraints and crowd out real investment. Externally, macroeconomic conditions are key determinants of liquidity. During monetary tightening cycles, firms with higher liquidity are better positioned to reduce agency conflicts. Conversely, in expansionary phases, the cost of idle funds may erode firm value. Firms facing tighter financing constraints tend to adjust their liquidity strategies more rapidly Meng et al. (2018). J. Zhang (2020)concluded that monetary easing is negatively correlated with corporate liquidity; however, more constrained firms may still choose to increase their cash ratios.

Some studies specifically examine how particular policies influence corporate liquidity. H. Ma (2024) argued that sufficient liquidity contributes to macroeconomic stability, but warned of the potential for financial risks if excess liquidity flows into speculative sectors and undermines the real economy. Q. Zhang and Huang (2019) proposed that a firm's soft power can stabilize stakeholder relationships and thus improve liquidity constraints. Ni et al. (2024) further emphasized that adequate liquidity supports real economic stability, though over-financialization may diminish its value and increase systemic risk.

Overall, existing literature on tax incentives, VAT credit refund policies, and corporate liquidity is relatively mature and multidimensional. Most empirical studies rely on data from publicly listed companies due to availability, though a few—such as X. Cui et al. (2023) - have used national tax survey data to explore employment effects. Dependent variables in the literature span macroeconomic, firm-level, and managerial dimensions, and many studies recognize the heterogeneous effects of tax policies across firm size, industry, and ownership structure. However, prior studies on VAT credit refund policy rarely examine firms' comprehensive financial condition as a primary outcome. Instead, they often focus on individual aspects of financial behavior or single performance indicators. This paper aims to

address that gap by using corporate financial resilience—defined as a firm's ability to withstand external shocks—as the core dependent variable. This variable is constructed from multiple indicators (as detailed in Section 4), allowing for a more accurate reflection of how firm financial conditions evolve in response to policy implementation and over time.

3 Theoretical Hypotheses

The VAT credit refund policy can enhance a firm's cash flow and reduce its outstanding debt, thereby improving both cash flexibility and debt flexibility, and ultimately strengthening overall financial resilience. Managers often prefer to rely on internal funding to maintain financial flexibility. This strategy enables them to effectively control and manage corporate assets and improve operational performance. During periods of economic downturn—such as the COVID-19 pandemic—additional cash flows can serve as a financial buffer, helping firms to sustain operations, pay employee wages, and manage revenue shortfalls (Fahlenbrach et al. (2020)). This risk-buffering capacity can prevent firms from becoming insolvent and being forced into restructuring or asset liquidation.

With the inflow of additional liquidity, firms reduce their reliance on debt financing, leading to lower interest expenses. Moreover, they can allocate freed-up capital to other business areas, which is conducive to firm growth and expansion. The strengthened financial position and reduced dependence on external debt improve firms' overall financial resilience. Based on this reasoning, we propose hypothesis H_1.

H1: The VAT credit refund policy significantly enhances corporate financial resilience.

Whereas most existing literature has focused on average treatment effects of the VAT refund policy, this paper intends to analyze the temporal heterogeneity and long-term effects by decomposing the policy's influence over time. We argue that corporate liquidity plays a mediating role in the policy's impact on financial resilience. This mechanism is supported by three theoretical frameworks: Signaling Theory, Corporate Life Cycle Theory, and Financial Constraint Theory.

I. Signaling Theory. Firms convey information about their quality and future prospects through financial decisions (Spence (1973)). Under the VAT refund policy, firms may prioritize the use of additional liquidity to repay debt and improve their credit standing. By doing so, they send a positive signal to creditors, investors, and other stakeholders regarding their financial stability and repayment capacity. This behavior reduces default risk, improves credit ratings, and lowers future financing costs (Ross (1977)). Moreover, firms may deliberately limit investment and R&D spending in the short run to focus on credit restoration, aligning with signaling theory's view that sacrificing short-term profits can enhance long-term credibility (Connelly et al. (2011)).

II. Corporate Life Cycle Theory. Financial and investment decisions vary depending on the firm's development stage (D. Miller and Friesen (1984)). Growth-stage firms often face tighter financial constraints and are more burdened by debt, limiting their investment capacity. Repaying debt to reduce financial risk is a strategic move that supports long-term development (Dickinson (2011)). The redirection of funds from debt repayment toward investment and R&D after an initial deleveraging phase indicates a transition from financial repair to expansion. Empirical studies, such as Dickinson (2011)), confirm that growth-stage firms often optimize capital structures (e.g., by reducing leverage) to accumulate resources for subsequent innovation and investment.

III. Financial Constraint Theory. It argues that due to information asymmetry, agency problems, and market frictions, the cost of external financing is significantly higher than internal funds. As a result, firms face investment constraints (Fazzari et al. (1988)). Under perfect market assumptions, financing

choices should be neutral to firm value (Modigliani and Miller (1958)), but real-world imperfections challenge this notion.The theory includes two notable sub-theories: (1) Pecking Order Theory (Myers and Majluf (1984)), which suggests that firms prefer internal funding over debt, and debt over equity, due to asymmetric information; (2) Agency Theory (Jensen and Meckling (1976)), which proposes that agency conflicts between managers and investors raise financing costs and amplify constraints. We infer the following policy impact pathway: The VAT credit refund provides additional liquidity, which firms first use to reduce short-term debt, lower interest expenses, and mitigate financial risk. This improves short-term indicators such as the current and quick ratios. These improvements signal financial health to banks and investors, potentially increasing access to external capital—consistent with signaling theory. As the firm moves from deleveraging to long-term profitability goals, the refunded capital facilitates strategic investment. In this context, improved financial resilience reflects not only enhanced liquidity management but also resource accumulation to withstand future shocks and maintain diversified financing channels. Thus, we propose the hypothesis H_2.

H2: The VAT credit refund policy has a sustained effect over time, initially reducing current liabilities through increased liquidity, and subsequently promoting investment expansion.

Furthermore, we explore the heterogeneous policy effects across different types of firms. Compared to large firms, small firms typically face higher financing costs and more severe information asymmetries. Myers and Majluf (1984) note that small firms rely more heavily on internal capital. The VAT refund increases internal liquidity, thus easing financing constraints and reducing reliance on debt. Fazzari et al. (1988) emphasize that financially constrained firms are more susceptible to financial distress, and tax incentives can mitigate default risk and improve financial flexibility. Additionally, capital-intensive firms require large upfront investments and face long payback periods. The refund policy extends the duration of usable capital and relieves liquidity pressure (Ohrn (2019)). It can also foster equipment upgrades and technological innovation, which indirectly strengthen financial resilience (Feng et al. (2024)). Therefore, we propose three subgroup hypotheses H_3a, H_3b, and H_3c **H3a: The VAT credit refund policy has a greater impact on financial resilience for firms facing higher financing constraints. H3b: The VAT credit refund policy has a greater impact on financial resilience for smaller firms. H3c: The VAT credit refund policy has a greater impact on financial resilience for capital-intensive firms.**

Taken together, the VAT Credit Refund's impact on corporate financial flexibility and its mechanism are summarized in Figure 3.1. As shown, the policy supplies firms with additional liquidity, thereby significantly enhancing their financial resilience, especially by improving their cash flexibility.

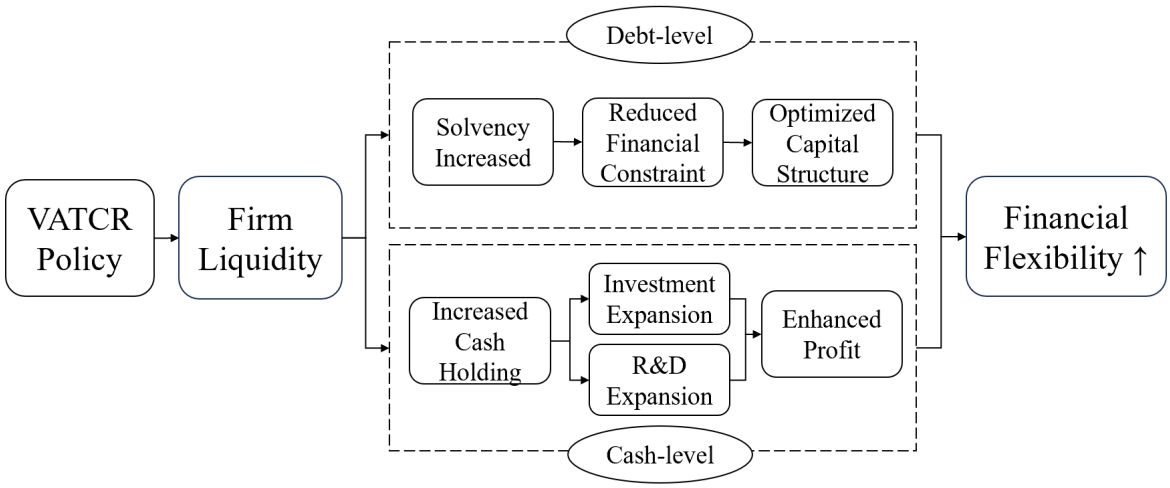

Figure 3.1: Mechanism

4 Basic Empirical Analysis

4.1 Variable Selection and Definitions

I. Dependent Variables

The primary dependent variable in this study is the firm Financial Flexibility index. Given the limited literature on the construction of this index, we build upon existing frameworks to develop a composite measure. Following C. Ma (2010), we classify financial resilience into two first-level indicators and six second-level indicators, aggregated using weighted sums (see Table 4.1). First-level indicator weights are determined via Analytic Hierarchy Process (For detailed calculation process, see Appendix A). The second-level weights are generated by using the coefficient of variation method[6]. The calculated weights is shown in Table 4.2.

[6]For inverse indicators, including Cash Flow Volatility, Short-term Debt Ratio, Constrained Debt Ratio, we adjusted their direction for consistency. Following C. Ma (2010), higher Cash Flow Volatility typically indicates lower resilience; thus, we use the reciprocal. Similarly, short-term and constrained debt ratios are transformed to "1 - ratio" to align them. Finally, all values are normalized to 0–1 scale using min-max transformation before computing their coefficients of variation.

Table 4.1: Indicators Related to Financial Resilience

First-Level Indicator	Second-Level Indicator	Formula / Description	Relationship to Resilience
Debt-side	Potential Debt Capacity	1 – Debt-to-Asset Ratio	Positive
	Short-Term Debt Ratio	Current Liabilities / Total Liabilities	Negative
	Constrained Debt Ratio	(Accounts Payable + Taxes Payable) / Total Liabilities	Negative
Cash-side	Cash Holdings	Cash & Equivalents / Total Assets	Positive
	Cash Flow Volatility	Std. Dev. of Operating Cash Flows	Negative
	Cash Output Capacity	Net Income / Total Assets	Positive
	Net Cash Surplus	(Cash – Current Liabilities) / Total Assets	Positive
External Financing Cost	Z-score	Altman Z-score	Positive

Table 4.2: Indicators' Weights

First-Level Indicator	Second-Level Indicator	Formula	Relationship to Resilience
Debt-side	0.650	Potential Debt Capacity	0.485
		Short-Term Debt Ratio	0.447
		Constrained Debt Ratio	0.068
Cash-side	0.290	Cash Holdings	0.298
		Cash Flow Volatility	0.068
		Net Cash Surplus	0.080
External Financing Cost	0.060	Altman Z-score	-

II. Independent Variables

This study treats the VAT credit refund policy introduced in 2018 and expanded in 2022 as exogenous shocks to construct a multi-period Difference-in-Differences (DID) model. In this study, we construct a multi-period Difference-in-Differences (DID) model using two key dummy variables to identify the policy treatment effect. The time dummy variable ("Post") takes the value of 1 for firms in years 2018 and 2022[7] onwards, corresponding to the implementation and expansion phases of the VAT credit refund policy, and 0 for all other years. The treatment dummy variable ("Treat") equals 1 for firms belonging to industries explicitly included in the 2018 pilot program—defined in MOF Document No.70, and for those newly covered under the 2022 policy expansion, while firms outside these designated sectors are assigned a value of 0. The interaction term between these two dummies captures the causal impact of the VAT refund policy on corporate financial resilience.

[7]Although the policy was expanded further in 2019, eligibility criteria are difficult to observe in firm-level data. Specifically, firm whose incremental credit accumulation over six consecutive months exceeding RMB 500,000 (USD $69,253) can be favored by the policy. Therefore, 2019 is not selected as a single treatment year.

4.2 Model Specification

This paper employs panel data to construct a multi-period Difference-in-Differences (DID) model for empirical analysis. To isolate the causal effect of the VAT credit refund policy on corporate financial resilience, we incorporate a range of control variables and introduce fixed effects at the firm level, year level, and province-by-year level. These adjustments account for unobserved heterogeneity across firms, macroeconomic fluctuations over time, and region-specific variations in policy enforcement. The empirical model is specified as follows:

$$FF_{it} = \beta_0 + \beta_1 DID_{it} + \beta_2 X_{it} + \delta_i + \gamma_t + (p_i \times \gamma_t) + \epsilon_{it} \tag{2}$$

In this equation, FF_{it} represents the financial flexibility index of firm i in year t, serving as the dependent variable. The key explanatory variable,DID_{it}, is an interaction term combining the treatment group identifier and the post-policy implementation time dummy. The term X_{it} denotes a vector of firm-level control variables as described in Section 4.1. The fixed effects δ_i, γ_t, and $p_i * \gamma_t$ control for firm-specific, time-specific, and the cross of province-specific and time-varying factors[8], respectively. The error term ϵ_{it} captures residual unobserved influences. The coefficient of interest is β_1, which measures the average treatment effect of the policy. A significantly positive estimate of β_1 would indicate that the VAT credit refund policy has a beneficial effect on firms' financial resilience, thereby providing empirical support for Hypothesis H1. It is worth noting that in the post-reform Chinese tax system, provincial tax authorities typically coordinate tax enforcement within their jurisdiction. Since local tax bureaus at the city and county levels follow standardized administrative protocols and digital infrastructure prescribed by provincial-level tax authorities, including province-by-year fixed effects effectively captures variation in local enforcement intensity and administrative implementation.

4.3 Data Source and Descriptive Statistics

The empirical analysis is based on an unbalanced panel dataset comprising all A-share listed firms in China from 2015 to 2023. The primary data sources are the CSMAR and Wind[9] databases. To ensure data consistency and reliability, we conduct a series of data cleaning procedures. Specifically, we exclude financial sector firms due to their unique regulatory environment and balance sheet structure, as well as firms marked with ST* [10]. Observations with missing values for key variables are also deleted, along with firms reporting negative return on equity (ROE). Finally, we apply 2% winsorization at both tails to mitigate the disturbance of the outliers. After cleaning, the final dataset includes 11,624 observations, covering 1,951 firms.

Descriptive statistics are presented in Table 4.3. The financial flexibility index, the study's primary dependent variable, ranges from 7.94 to 12.28 with a mean of 10.31, reflecting substantial variation in firms' financial health. Firm size, measured as the log of total assets, ranges from 2.04 to 2.69. The return on assets (ROA) varies between 0 and 0.60, indicating large differences in profitability across firms. The

[8]After the merger of national and local tax administrations in China in 2018, tax collection and enforcement have generally been coordinated at the provincial level. Local tax bureaus at the municipal or county level operate under the direct supervision of their respective provincial tax authorities. As a result, within each province, tax enforcement procedures, the level of digitization, and the intensity of policy implementation tend to exhibit a high degree of consistency. Therefore, this paper includes province-by-year fixed effects to control for variations in provincial tax administration policies or enforcement strength over time and to mitigate potential biases arising from intra-provincial differences in tax enforcement. Official announcement of this revolution can be seen on https://www.gov.cn/xinwen/2018-07/20/content_5308075.htm.

[9]The CSMAR (China Stock Market & Accounting Research, see https://data.csmar.com/) and WIND (Wande) database offers comprehensive Chinese financial and economic data, including stock market, financial statements of listed companies, and corporate governance information. It's widely used for empirical research on China's capital markets, similar to CRSP/Compustat in the U.S. Super clutch for academic and investment analysis.

[10]Firms with ST* typically signals financial distress or abnormal operations, so we must delete them before regression.

average level of executive education is approximately 3.67, corresponding to a level between undergraduate and master's degrees. To improve the interpretability of regression coefficients, we multiply the financial flexibility index by a factor of 10.

Table 4.3: Descriptive Statistics

Variable	Obs.	Mean	SD	Min	Max
FF	11624	10.3125	0.5273	7.9462	12.2815
Boardsize	11623	2.1431	0.1989	1.0986	2.8332
Size	11624	2.2846	0.1316	2.0431	2.6880
Indp	11623	0.3764	0.0581	0.1667	0.8000
Dual	11624	0.2221	0.4170	0.0000	1.0000
Salegr	11624	0.1439	0.3147	-0.4376	1.8738
ROA	11624	0.0664	0.0554	-0.0869	0.3799
Tobing	11624	2.0312	1.3868	0.8004	8.7580
Degree	9901	3.6695	0.4819	2.6467	7.0000

4.4 Baseline Regression Results

This section presents the baseline estimation results from the multi-period Difference-in-Differences (DID) model, examining the effects of the 2018 and 2022 VAT credit refund policy shocks on corporate financial resilience. The regression results are reported in Table 4.4. Column (1) shows the results without including any control variables or fixed effects. In this simple specification, the VAT credit refund policy is associated with an average increase of 19.65% in the financial flexibility index, and the coefficient is highly significant. Column (2) adds firm, year, and province-by-year fixed effects. After controlling for these unobserved heterogeneity, the estimated policy effect remains positive and statistically significant at the 5% level, with an average increase of 3.17%. Columns (3) and (4) sequentially introduce control variables such as firm size, board structure, profitability, growth rate, executive education level, and Tobin Q. Column (4), which includes both the full set of controls and fixed effects, provides the most comprehensive specification. The estimated coefficient on the DID term in this model indicates that, relative to firms not affected by the policy, treated firms experienced an average increase of 2.63% in financial flexibility following the implementation of the VAT credit refund policy. This effect is statistically significant at the 5% level. These findings provide robust empirical support for Hypothesis H1, which posits that the VAT credit refund policy significantly enhances firms' financial resilience.

5 Robustness Tests

5.1 Parallel Trend Test

To verify the robustness of the multi-period DID regression results, we conduct a parallel trend test, taking the year immediately prior to the policy implementation (i.e. 2017 and 2022) as the reference period. The result shown in Figure 5.1. Compared to the conventional DID design, the multi-period DID test places more emphasis on the coefficients relative to the policy implementation year rather than on specific calendar years. For enterprises affected by the policy expansion in 2022, the earliest sample year (the year of 2015) is labeled as pre_7. Therefore, the parallel trend test spans from pre_7 to $post_5$. In practice, all years earlier than pre_5 are grouped into a single period labeled pre_5 to avoid sparse data issues. As shown in Figure 5.1, prior to the implementation of the policy, the difference between the treatment and control groups is statistically indistinguishable from zero, indicating the absence of a

Table 4.4: Baseline Regression Result

	(1) FF	(2) FF	(3) FF	(4) FF
DID	0.1965***	0.0317**	0.0513***	0.0263**
	(0.0092)	(0.0134)	(0.0087)	(0.0130)
Boardsize			-0.0305	-0.0503
			(0.0275)	(0.0327)
Size			3.6741***	3.6647***
			(0.1300)	(0.1444)
Indp			0.0783	0.0867
			(0.0787)	(0.0817)
Dual			-0.0147	-0.0140
			(0.0088)	(0.0094)
Salegr			-0.0347***	-0.0222**
			(0.0085)	(0.0084)
ROA			0.0181	0.0076
			(0.0768)	(0.0777)
Tobinq			0.0055**	0.0027
			(0.0033)	(0.0040)
Degree			0.0180	
			(0.0112)	
Constant	10.2189***	10.1588***	1.8613***	2.0354***
	(0.0124)	(0.0063)	(1.2730)	(0.3098)
Individual FE		✓		✓
Time FE		✓		✓
Province*Time FE		✓		✓
No. of Obs	11624	11624	9570	9570
Within-R^2	0.0828	0.2387	0.4090	0.4254

pre-trend. After the policy is introduced, the coefficients become significantly positive, confirming that the parallel trends assumption holds and supporting the validity of our DID identification strategy.

5.2 PSM-DID Test

The implementation of the VAT credit refund policy is subject to certain eligibility requirements, namely that firms must have outstanding VAT credit balances and maintain a good tax credit rating. The presence of VAT credit implies that the firm's input VAT exceeds its output VAT, while a good credit rating indicates a relatively healthy financial status. These conditions suggest the potential for self-selection bias: firms that qualify for the refund may already possess characteristics—such as stronger financial flexibility—that differentiate them from non-qualifying firms. To address this concern and improve the robustness of the causal inference, we apply Propensity Score Matching (PSM) in combination with the DID approach. Using firm-level control variables, we first estimate the probability that a firm receives the policy treatment, i.e., becomes eligible for the VAT credit refund. Based on these propensity scores, we perform two types of matching: nearest-neighbor matching and kernel matching. For each treated firm, we identify a control firm with the most similar propensity score. Firms that cannot be successfully matched are excluded from the subsequent analysis to avoid introducing imbalance. The nearest-neighbor matching yields 8,850 observations, while kernel matching results in 9,565 observations.

Figure illustrates that after matching, the standardized differences in all control variables fall below 10%, indicating that the distribution of covariates between the treatment and control groups has become well-balanced. Figure 5.2 and 5.3 presents the results of the common support test, which confirms

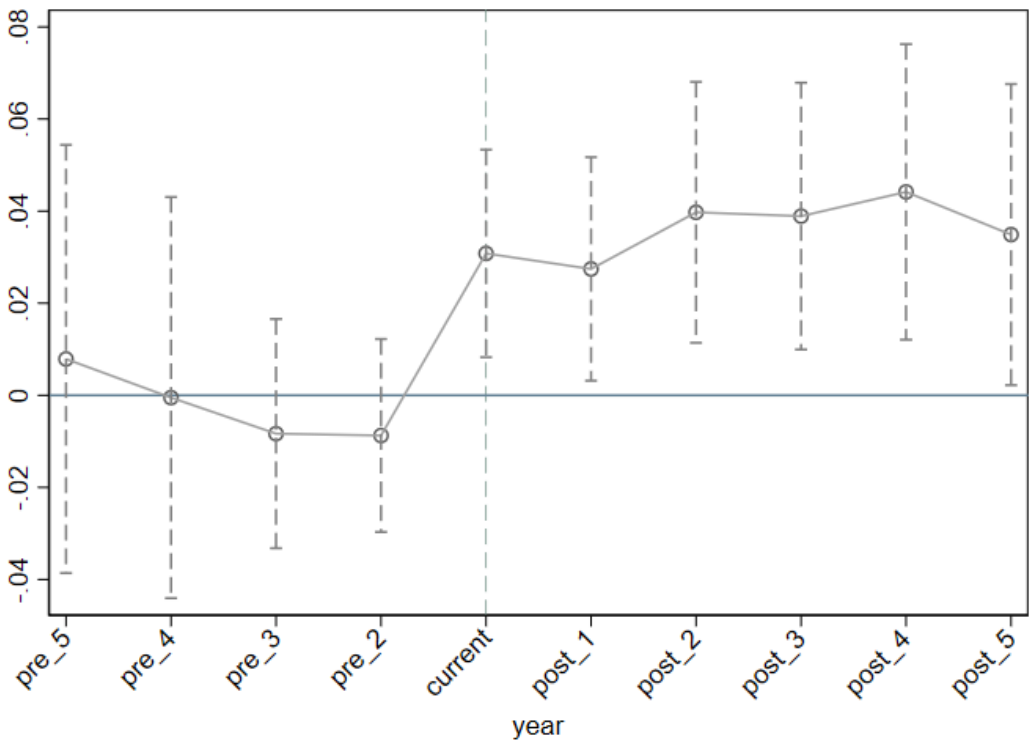

Figure 5.1: Parallel Trend Test

that treated and control firms share sufficient overlap in their propensity scores to justify comparison. Then, we re-estimate the DID model using the matched samples. The estimation results are reported in Table 5.1. Columns (1) and (2) display the nearest-neighbor matching results, while Columns (3) and (4) report the kernel matching results. Both approaches produce consistent findings: the VAT credit refund policy leads to an average increase in financial flexibility of 2.31% and 2.61%, respectively. These effects are statistically significant at the 5% level. The consistency across matching techniques reinforces the robustness of the baseline results and confirms that the observed policy effects are not driven by pre-existing differences between treated and control firms.

5.3 Placebo Test

To further verify the robustness of the results and rule out the possibility that the observed policy effects are driven by spurious time trends or coincidental shocks, this study conducts a placebo test. Specifically, we artificially assign the policy intervention to a period before the actual implementation and re-estimate the model to observe whether a significant treatment effect still emerges. If the DID coefficient remains insignificant under these falsified policy timings, it would support the argument that the true policy effect observed earlier is not due to random chance or time-varying unobservables. In practice, we randomly select a group of firms to act as the "placebo treatment group" and assign them a fake policy shock year, such as 2016 or 2017—years in which no VAT credit refund policy was implemented. We then replicate the baseline regression using this pseudo-treatment design. The results show that the estimated coefficients on the placebo DID terms are statistically insignificant, and their magnitudes are close to zero. This outcome confirms that the observed effect in the actual analysis is unlikely to result from pre-existing trends or unrelated concurrent policy shifts, thereby reinforcing the credibility of the causal interpretation of the VAT credit refund policy.

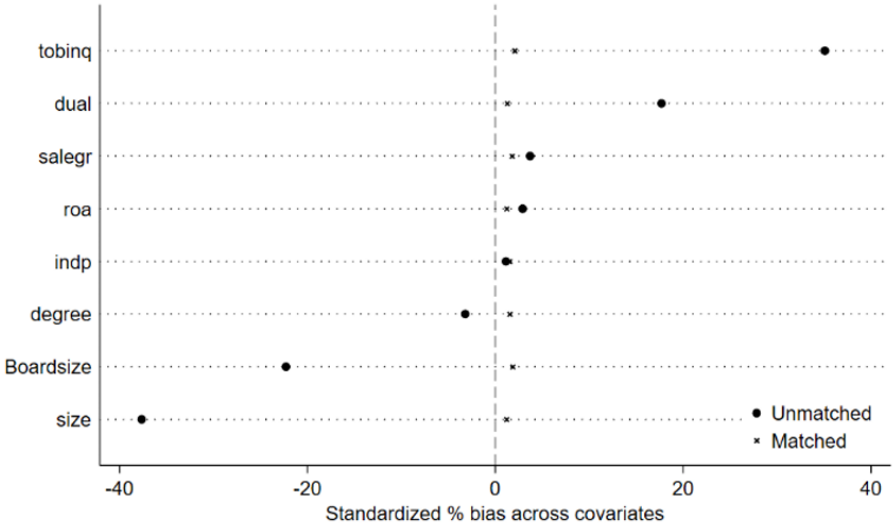

Figure 5.2: PSM-DID Nearest-Neighbor Matching Results

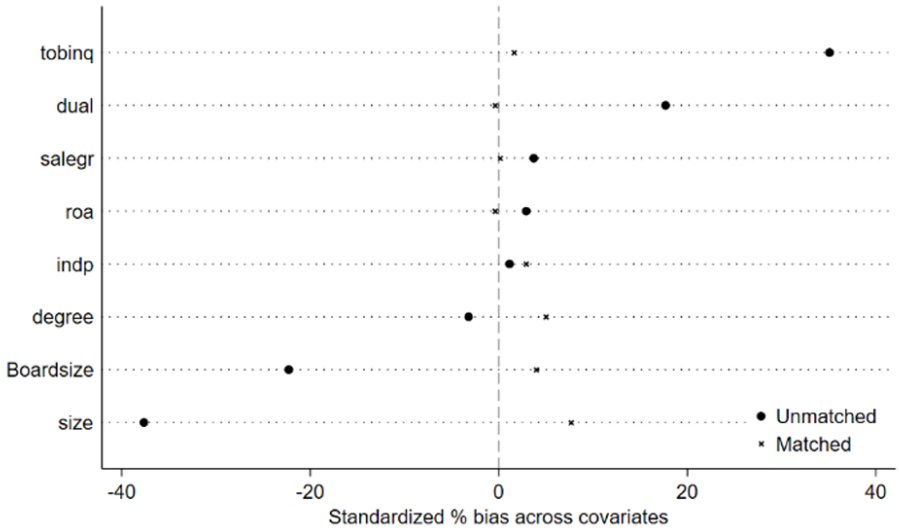

Figure 5.3: PSM-DID Kernel Matching Results

Table 5.1: PSM-DID Test Result

	NN Matching		Kernel Matching	
	(1) FF	(2) FF	(3) FF	(4) FF
DID	0.0489***	0.0231**	0.0510***	0.0261**
	(0.0087)	(0.0114)	(0.0087)	(0.0114)
Boardsize	-0.0225	-0.0459	-0.0279	-0.0481
	(0.0284)	(0.0356)	(0.0274)	(0.0341)
Size	3.6561***	3.6642***	3.6726***	3.6733***
	(0.0583)	(0.1325)	(0.0534)	(0.1260)
Indp	0.1093	0.0987	0.0789	0.0211
	(0.0818)	(0.0920)	(0.0779)	(0.0891)
Dual	-0.0131*	-0.0130	-0.0149*	-0.0141
	(0.0090)	(0.0110)	(0.0087)	(0.0098)
Salerg	-0.0360**	-0.0215**	-0.0347**	-0.0210**
	(0.0097)	(0.0091)	(0.0086)	(0.0091)
ROA	0.0550	0.0314	0.0177	0.0049
	(0.0532)	(0.0617)	(0.0493)	(0.0590)
Tobinq	0.0092	-0.0011	0.0058*	0.0018
	(0.0034)	(0.0045)	(0.0033)	(0.0043)
Degree	0.0187		0.0178	
	(0.1309)		(0.1272)	
Constant	1.8702**	2.0035***	1.8558***	2.0106***
	(0.1309)	(0.3131)	(0.1272)	(0.2993)
Individual FE		✓		✓
Time FE		✓		✓
Province*Time FE		✓		✓
No. of Obs	8850	8850	9565	9565
Within-R^2	0.4115	0.4246	0.4907	0.4260

5.4 Controlling for Concurrent Tax Policy Interventions

To ensure that the estimated effect of the VAT credit refund policy is not confounded by other concurrent tax reduction policies implemented during the same period, this section introduces additional robustness checks by isolating the influence of overlapping fiscal measures. In recent years, the Chinese government has launched several large-scale tax reduction initiatives, such as adjustments to the value-added tax rates, increases in the threshold for small-scale taxpayers, and reductions in corporate income tax for micro and small enterprises. These policies may also affect firms' liquidity and financial flexibility, posing a potential threat to the identification strategy if not properly controlled. To address this concern, we refine the sample and model in two ways. First, we exclude firms from industries and regions that were simultaneously affected by other major tax policy changes. Second, we incorporate policy dummy variables for key concurrent reforms to account for their potential influence. After applying these adjustments, the DID coefficient associated with the VAT credit refund policy remains positive and statistically significant, and its magnitude is only slightly attenuated compared to the baseline estimate. These results suggest that the observed improvement in financial flexibility is indeed attributable to the VAT credit refund policy itself, rather than being driven by other tax incentives introduced during the same period. This finding strengthens the internal validity of the study and confirms that the policy's effect is not spuriously inflated due to unaccounted external tax shocks.

For more robustness checks, such as altering fixed effects level, see Appendix B, in terms of changing different estimators clustering strategies, see Appendix C.

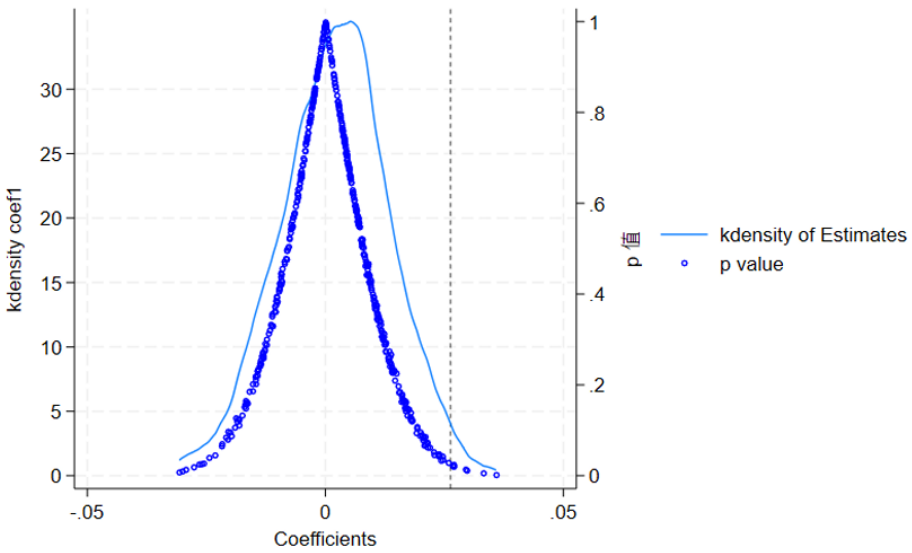

Figure 5.4: Placebo Test

Table 5.2: Controlling for Concurrent Tax Policy Interventions

	(1) FF	(2) FF	(3) FF	(4) FF
DID	0.1961***	0.0284**	0.0487***	0.0231**
	(0.0092)	(0.0122)	(0.0087)	(0.0116)
TaxRate	-0.0023***	-0.0021**	-0.0023***	-0.0019***
	(0.0007)	(0.0008)	(0.0008)	(0.0008)
Controls			✓	✓
Individual FE		✓		✓
Time FE		✓		✓
Province*Time FE		✓		✓
No. of Obs	11226	11226	9229	9229
Within-R^2	0.0827	0.2613	0.4128	0.4465

6 Further Analysis

To more precisely understand how the VAT credit refund policy enhances firms' financial resilience, this section conducts an in-depth analysis from three perspectives: mechanism identification, dynamic effects, and heterogeneity analysis. Specifically, we investigate the transmission mechanism through which the policy affects financial resilience, examine the temporal evolution of policy effects, and explore whether the impact varies across firms with different characteristics.

6.1 VAT Credit Refund and Corporate Liquidity

Corporate liquidity reflects a firm's ability to meet short-term financial obligations, which is typically measured by the relationship between available cash flows and outstanding short-term liabilities. Existing literature suggests that firms with abundant cash flows and relatively low short-term debt burdens exhibit higher liquidity, and thus greater financial flexibility when facing external shocks or investment opportunities (Acharya et al. (2007); Almeida et al. (2004); Faulkender and Wang (2006)). Acharya et al. (2007) argue that corporate cash holdings serve as a hedge against debt-related risks: firms with stronger cash positions are better able to withstand liquidity shocks arising from short-term liabilities, thereby

demonstrating higher liquidity. Almeida et al. (2004) examine the relationship between cash-flow sensitivity and cash holdings, finding that firms with stronger cash flows tend to accumulate reserves to buffer against potential future financing constraints or debt repayment pressures, thereby improving liquidity. Faulkender and Wang (2006) further show that variations in corporate cash flows significantly affect firm value, partly because cash flows directly reflect repayment capacity when short-term debt matures. Firms with stronger cash flows and lower proportions of short-term debt generally demonstrate greater repayment ability, which enhances their valuation in capital markets and signals investors' recognition of their liquidity.

Drawing on this literature, we select the following indicators as mechanism variables to measure corporate liquidity, which are incorporated as dependent and control variables in the regressions (see Table 6.1). From the liability perspective, two variables are considered: (1) the inverse of the current ratio (current liabilities divided by current assets, DEBT) to capture short-term debt structure; and (2) the leverage ratio (LEV) to capture the overall debt structure, including both short- and long-term liabilities. The VAT Credit Refund policy increases firms' free cash flow, significantly strengthening short-term debt-servicing capacity and reducing reliance on costly external financing. From the cash-flow perspective, we adopt (1) the ratio of net operating cash flow to current liabilities (CFr) to measure cash repayment capacity, i.e., the ability to service short-term debt using cash flows; and (2) "tax rebates received + other operating cash inflows" (TaxRebate), log-normalized, to capture liquidity from the operational side (Wang Min et al., 2024). These two measures reflect firms' enhanced short-term debt-servicing capacity and their ability to mobilize sufficient cash in response to economic fluctuations, thereby highlighting financial flexibility and resilience. To capture financing constraints, we employ the SA and KZ indices (Hadlock and Pierce (2010); Kaplan and Zingales (1997)), where higher SA or KZ values indicate stronger financing constraints.

Table 6.1 reports the regression results. Column (4) shows that the VAT Credit Refund policy has a significantly positive effect on firms' cash repayment capacity, with coefficients significant at the 5% level. Column (2) demonstrates that the policy significantly reduces current liabilities, thereby improving the short-term debt structure, suggesting that firms use refund proceeds to repay short-term obligations. Column (3) reports insignificant coefficients, indicating that the policy does not significantly affect the actual tax rebate received. Columns (5) and (6) show significantly negative coefficients, suggesting that financing constraints are alleviated under the policy shock.

Table 6.1: Mechanisms

	(1) LEV	(2) DEBT	(3) TaxRebate	(4) CFr	(5) KZ	(6) SA
DID	-0.0068**	-0.2397**	0.0001	0.0220**	-0.1124**	-0.0093**
	(0.0027)	(0.1068)	(0.0363)	(0.0112)	(0.0543)	(0.0029)
Controls	✓	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓
Province*Time FE	✓	✓	✓	✓	✓	✓
Observations	9570	9570	8081	9570	9570	9570
Within-R^2	0.1889	0.0317	0.4034	0.0722	0.0635	0.8629

6.2 Dynamic Effects

6.2.1 VAT Credit Refund and Firm Fund Utilization

Although Section 4 has demonstrated that the VAT Credit Refund policy helps enhance firms' financial flexibility, how firms actually utilize the refunded funds constitutes a crucial factor determining whether such effects can be sustained. To this end, we further examine changes in pilot firms' cash flows, cash holdings, debt repayment, and fixed asset investment following the implementation of the refund policy, thereby uncovering the financial behaviors underlying the allocation of refunded funds.

The analysis of fund utilization is divided into two dimensions: inflows and outflows. On the inflow side, we select the logarithm of net operating cash flow (CF) and the ratio of net operating cash flow to current liabilities (CFr, consistent with Section 6.1); the logarithm of cash holdings (CH), as well as the cash holdings ratio (CHr), measured by cash and cash equivalents and their proportion of total assets. On the outflow side, we consider fixed asset investment (IIN), measured by the ratio of net fixed assets to total assets; R&D expenditure (RD), measured as the ratio of R&D expenditure to operating revenue; and investment in acquired assets (IAA), defined as "cash payments for the purchase and construction of fixed assets, intangible assets, and other long-term assets + R&D expenditure" relative to operating revenue (Wang and Li (2024)). Table 6.2 reports the results: while firms' cash flows increased significantly, the overall effects on cash holdings, investment, and R&D expenditures are not statistically significant.

Table 6.2: VAT Credit Refund and Corporate Capital Utilization

	(1) CF	(2) CFr	(3) CH	(4) CHr	(5) IIN	(6) IAA	(7) RD
DID	0.0752*	0.0220**	-0.0102	-0.0007	0.0042	-0.0848	0.0022
	(0.0416)	(0.0112)	(0.0272)	(0.0011)	(0.0110)	(0.0835)	(0.0016)
Controls	✓	✓	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓
Province*Time FE	✓	✓	✓	✓	✓	✓	✓
Observations	8379	9570	9570	9570	9570	8473	8474
Adj-R^2	0.2322	0.0317	0.5530	0.0722	0.0307	0.7216	0.8629

We select the variables CFr, CHr, IIN, IAA, and RD from Table 6.2 and employ an event-study approach to conduct dynamic effect analyses, retaining and displaying the sample years from the year of policy implementation onward. The results, as reported in Table 6.2.1, reveal the following:
First, firms' cash-flow repayment capacity significantly increases (Column 1), but the effect is heterogeneous over time. It is significant in the policy implementation year and up to three years prior, but becomes insignificant in later years. 2. Cash holdings significantly decrease starting from the third year after the policy implementation (Column 2), consistent with the findings of Xie et al. (2022). In addition, current liabilities significantly decrease (Column 3): firms' current liabilities decline markedly in the first year after the policy implementation, with the effect persisting until the penultimate sample year. Fourth, the overall effect on investment is insignificant (Column 4), though it becomes significant from the third year onward. In contrast, investment in acquired assets shows no significant increase in any sample year (Column 5). Finally, the total effect on R&D expenditure is insignificant (Column 6), but becomes significant starting two years after the policy shock.

Based on these findings, the following conclusions can be drawn: To begin with, upon receiving VAT refunds, firms prioritize using additional cash flows to alleviate short-term financial pressures rather than

Table 6.3: Dynamic Effect

	CFr	CHr	DEBT	IIN	IAA	RD
Total Effect	0.0220**	-0.0007	-0.2397**	0.0042	-0.0848	0.0022
	(0.0112)	(0.0012)	(0.1068)	(0.0110)	(0.0835)	(0.0016)
Current	0.0236*	-0.0019	-0.0466	0.0187	-0.1826	-0.0020
	(0.0111)	(0.0012)	(0.0664)	(0.0138)	(0.1930)	(0.0033)
$Post_1$	0.0152	0.0016	-0.1697*	0.0045	-0.2104	-0.0002
	(0.0163)	(0.0015)	(0.0911)	(0.0160)	(0.2062)	(0.0034)
$Post_2$	0.0276*	0.0004	-0.1933*	0.0130	-0.2334	0.0048**
	(0.0155)	(0.0017)	(0.1063)	(0.0270)	(0.2390)	(0.0023)
$Post_3$	0.0272*	-0.0030*	-0.2141***	0.0259*	-0.1602	0.0035
	(0.0158)	(0.0017)	(0.0579)	(0.0141)	(0.1551)	(0.0023)
$Post_4$	0.0179	-0.0027	-0.2953***	0.0219**	-0.1838	0.0049**
	(0.0194)	(0.0019)	(0.0835)	(0.0098)	(0.1839)	(0.0024)
$Post_5$	0.0197	-0.0043**	-0.1985	0.0166*	-0.1899	0.0084***
	(0.0139)	(0.0021)	(0.0951)	(0.0089)	(0.1734)	(0.0027)
Controls	✓	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Province*Year FE	✓	✓	✓	✓	✓	✓
No. of Obs	9570	9570	9570	9570	8473	8474
Adj-R^2	0.0740	0.1158	0.1808	0.4487	0.0180	0.0630

channeling them directly into high-risk, long-horizon investment projects. Both Tables 6.2 and 6.3 show that VAT refunds significantly increase firms' cash flows, which are then primarily used to repay current liabilities and reduce financial risks. This suggests that firms prefer to use internal funds for debt repayment to optimize capital structure, thereby laying the groundwork for subsequent investment and R&D activities. Second, the VAT Credit Refund policy increases firms' fixed asset investment from the third year onward, but does not significantly enhance investment in acquired assets. This outcome reflects the industrial characteristics of beneficiary sectors, which are predominantly capital-intensive industries such as manufacturing. These industries tend to allocate funds toward equipment maintenance, upgrades, and environmental retrofitting, all of which are categorized as fixed asset investment but not as asset acquisitions. In the transmission process, VAT refunds also stimulate supply chain upgrading, indirectly encouraging suppliers to expand investment in specialized equipment. While this synergy boosts overall investment, it is not directly captured by firms' acquisition data. Lastly, the policy exerts a significant effect on R&D investment. Given the flexible, short-cycle, and technology-intensive nature of R&D, firms are more inclined to allocate refund funds to R&D rather than to fixed asset investment. Between 2020 and 2023, the pandemic narrowed sales channels, led to inventory backlogs, and weakened profitability, dampening firms' willingness to undertake fixed asset investment, especially asset acquisitions. Consequently, VAT refunds were primarily used to repay debt and to fund equipment maintenance and upgrades, which also bear R&D-like characteristics. Overall, the use of VAT refund funds exhibits temporal heterogeneity. Firms initially deploy refunds to repay short-term debt, mitigate financing constraints, and strengthen liquidity. Once liquidity improves, they gradually shift toward longer-term objectives, enhancing future profitability through investment and R&D. This sequential pattern aligns with the distinction between firms' short-term and long-term goals discussed earlier. By simultaneously reducing debt pressure and releasing liquidity, the VAT Credit Refund policy effectively enhances financial flexibility. However, the policy's effects are subject to temporal fluctuations. For instance, the marginal gains in financial flexibility diminish in the later years of implementation, likely due to changes in the external

economic environment or firms' internal capital allocation strategies. In 2023, as firms resumed production and VAT credit balances declined, the refunds they received diminished, and the enhancement of financial flexibility was no longer significant. The longer-term effects of the policy will require data from 2024 onward to be fully observed. To better illustrate the dynamic effects, the visualization of Table 6.2.1 is presented in Figure 6.1.

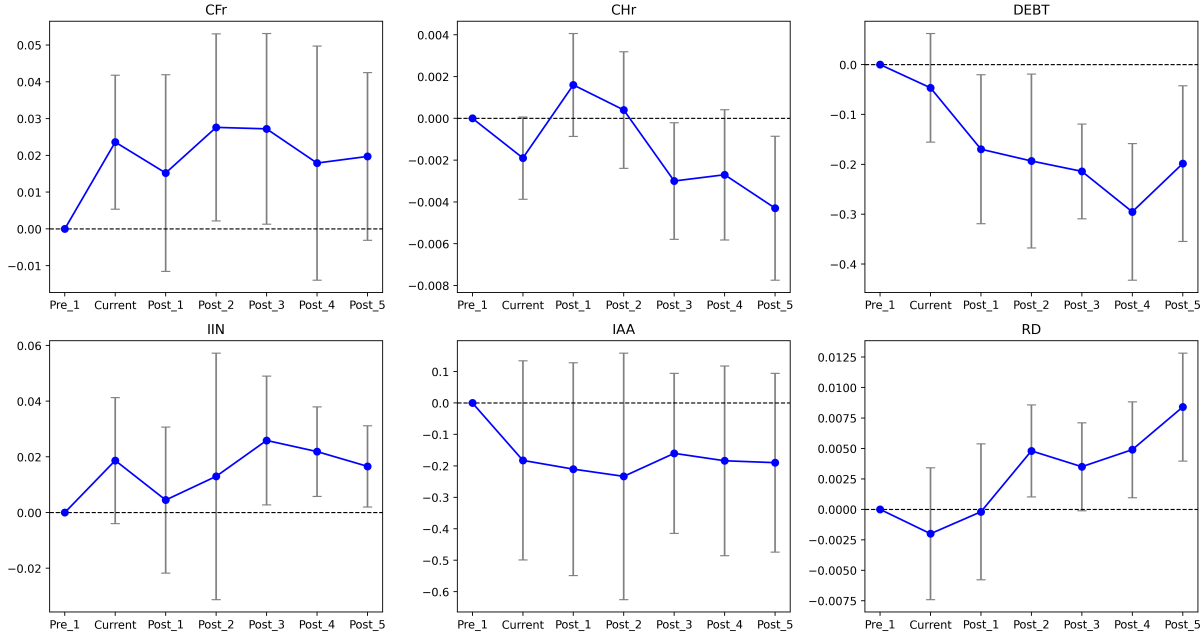

Figure 6.1: Dynamic Effect

6.3 Heterogeneity Analysis

To examine the differential impact of the VAT credit refund policy on firms with varying characteristics, this paper conducts a heterogeneity analysis by exploring the moderating effects of four firm-level variables: financial constraints, firm size, capital intensity (CI, measured as the ratio of fixed assets to total assets), and executive education level. Financial constraints are measured using the SA index, which is derived from firm size and firm age. To identify heterogeneity, firms in the top 20% of the SA index distribution—those facing the most severe financial constraints—are assigned a dummy variable, SA_High = 1; otherwise, the value is 0. For capital intensity and firm size, firms are divided into two equally sized groups based on the median value of each variable. Firms above the median are assigned dummy variables CI_High and Size_High, respectively, equal to 1; otherwise, they are assigned 0. As for executive education (Degree), firms with an average executive education level of undergraduate or above ($Degree \geq 4$) are assigned a value of 1, representing high education; otherwise, the value is 0. Finally, these dummy variables are interacted with the policy indicator variable DID to construct a heterogeneity analysis model, as shown in Equation 3.

$$FF_{it} = A_{it_High} \times DID_{i,t} + A_{it_High} + DID_{i,t} + X_{it} + \epsilon_{it} \tag{3}$$

The model focuses on the interaction term $A_{it}^{High} * DID$. A significantly positive coefficient on this term indicates that firms with higher values of the corresponding variable experience greater improvements in financial resilience, and vice versa. Table 6.4 presents the regression results. Specifically, the interaction term for financial constraints is significantly positive, suggesting that financially constrained

firms exhibit larger improvements in financial resilience. The interaction term for firm size is significantly negative, indicating that smaller firms benefit more from the policy. The coefficient for capital intensity is significantly positive, implying that capital-intensive sectors (e.g., various manufacturing industries) show more pronounced improvements in financial resilience. However, the interaction term for executive education level is not statistically significant, indicating no clear relationship between managerial education and the firm's capacity to leverage tax incentive policies. In addition, this paper applies alternative methods of heterogeneity analysis, including variables such as firm growth potential and financial soundness. The detailed procedures and results can refer to Appendix D.

Table 6.4: Heterogenous Analysis

	(1) FF	(2) FF	(3) FF	(4) FF
Size_High*DID	-0.0417**			
	(0.0092)			
CI_High*DID		0.0341*	-0.0305	-0.0503
		(0.0200)		
Degree_High*DID		-0.0062		
			(0.0191)	
SA_High*DID				0.0460**
				(0.0225)
DID	0.0503***	0.0230*	0.0274**	-0.0261**
	(0.0183)	(0.0123)	(0.0088)	(0.0225)
Individual FE	✓	✓	✓	✓
Time FE	✓	✓	✓	✓
Province*Time FE	✓	✓	✓	✓
No. of Obs	9570	9570	9570	9570
Within-R^2	0.4039	0.4040	0.4023	0.2379

7 Conclusion and Policy Implications

This paper investigates the impact of China's VAT credit refund policy on corporate financial resilience by employing a multi-period Difference-in-Differences (DID) framework, complemented by Propensity Score Matching (PSM) and a series of robustness checks. The findings indicate that the VAT credit refund policy significantly enhances the financial flexibility of treated firms, particularly by improving cash holdings and reducing reliance on short-term debt. This positive effect is robust across different model specifications and persists over time, with dynamic analysis revealing that the benefits accumulate in the years following policy implementation. Moreover, the policy's effects are more pronounced among private firms, small and medium-sized enterprises (SMEs), and firms operating in high-tech sectors.

These findings have important policy implications. First, liquidity-oriented tax relief policies, such as the VAT credit refund, can serve as an effective tool for stabilizing microeconomic fundamentals during periods of uncertainty or economic transition. By directly easing firms' financing constraints, such policies improve their capacity to withstand external shocks and maintain business continuity. Second, the heterogeneous effects observed across ownership types and industries suggest that precision targeting in fiscal policy design can enhance overall policy efficiency. Tailoring tax relief programs to the needs of financially constrained or innovation-intensive firms may yield greater economic resilience and promote high-quality development. Third, the findings also highlight the value of continued investment in tax administration reform and digital infrastructure at the provincial level, as consistent implementation

quality is essential to realizing the full benefits of tax policies.

Future research may explore the long-term consequences of VAT credit refunds on firm productivity, innovation, or employment, as well as investigate potential spillover effects along the supply chain. As China continues to refine its tax and fiscal framework, empirical evidence such as that provided in this paper can offer valuable insights for the formulation of more adaptive, inclusive, and growth-oriented policies.

Note on References[11]

References

Acharya, V. V., Almeida, H., & Campello, M. (2007). Is cash negative debt? a hedging perspective on corporate financial policies. *Journal of Financial Intermediation*, *16*(4), 515–554.

Almeida, H., Campello, M., & Weisbach, M. S. (2004). The cash flow sensitivity of cash. *The Journal of Finance*, *59*(4), 1777–1804.

Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, *5*(1), 31–56.

Bird, R., & Gendron, P. P. (2007). *The vat in developing and transitional countries*. Cambridge University Press.

Cailin, S., & Zidar, O. (2020). *Tax policy and the economy* (tech. rep.). NBER.

Connelly, B. L., Certo, S. T., Ireland, R. D., & Reutzel, C. R. (2011). Signaling theory: A review and assessment. *Journal of Management*, *37*(1), 39–67.

Cui, H., Tian, M., & Wang, Q. (2022). Does vat credit retention suppress corporate r&d investment? [In Chinese]. *Trade Economics*, *43*(8), 59–73.

Cui, X., Cai, Y., & Lu, G. (2023). Can VAT Refunds Promote Employment? Evidence from the 2019 Pilot Program [In Chinese]. *Management World*, *39*(9), 15–38.

De la Feria, R. (2018). Vat and the eu internal market: The challenges of harmonisation. *Intertax*, *46*(6–7), 489–502.

DeAngelo, H., & DeAngelo, L. (2007). *Capital structure, payout policy, and financial flexibility* [Working Paper, University of Southern California].

Desai, M. A., & Dharmapala, D. (2009). Corporate tax avoidance and firm value. *The Review of Economics and Statistics*, *91*(3), 537–546.

Desai, M. A., Goolsbee, A., & Glaeser, E. (2020). *Tax incentives and firm behavior during economic crises* (tech. rep.). NBER.

Dickinson, V. (2011). Cash flow patterns as a proxy for firm life cycle. *The Accounting Review*, *86*(6), 1969–1994.

Eljelly, A. M. (2004). Liquidity-profitability tradeoff: An empirical investigation in an emerging market. *International Journal of Commerce and Management*, *14*(2), 48–61.

Fahlenbrach, R., Rageth, K., & Stulz, R. M. (2020). How valuable is financial flexibility when revenue stops? evidence from the covid-19 crisis. *SSRN Electronic Journal*.

[11]Note on References: References of this paper includes both English and Chinese references. For clarity and ease of identification, all references originally published in Chinese are marked with "[In Chinese]" at the end of each citation. These sources primarily consist of academic journal articles, working papers, and government documents relevant to China's tax policies and corporate finance. The inclusion of Chinese literature enriches the contextual understanding of the VAT credit refund policy and its application in the Chinese institutional setting.

Faulkender, M., & Wang, R. (2006). Corporate financial policy and the value of cash. *The Journal of Finance*, *61*(4), 1957–1990.

Fazzari, S. M., Hubbard, R. G., & Petersen, B. C. (1988). Financing constraints and corporate investment. *Brookings Papers on Economic Activity*, *1988*(1), 141–206.

Feng, Q., Ge, Y., & Zhao, L. (2024). Tax incentives and corporate digital transformation: Evidence from china's accelerated depreciation policy. *Journal of Asian Economics*, *95*, 101832.

Ferris, J. S. (1981). A transactions theory of trade credit use. *The Quarterly Journal of Economics*, *96*(2), 243–270.

Gale, W. G., Gelfond, H., Krupkin, A., Mazur, M. J., & Toder, E. J. (2018). Effects of the tax cuts and jobs act: A preliminary analysis. *National Tax Journal*, *71*(4), 589–611.

Gamba, A., & Triantis, A. (2008). The value of financial flexibility. *Journal of Finance*, *63*(5), 2263–2296.

Gansu Provincial Tax Bureau Research Group, Yu, Y., Lyu, B., et al. (2024). The Incentive Effect of VAT Refunds on Enterprise Investment [In Chinese]. *Taxation Research*, (9), 114–121.

Gravelle, J. G. (2006). *Tax havens: International tax avoidance and evasion* (tech. rep.). Congressional Research Service.

Gravelle, J. G. (2014). *Corporate tax incidence: Review of general equilibrium estimates and analysis* (tech. rep.). Congressional Research Service.

Hadlock, C. J., & Pierce, J. R. (2010). New evidence on measuring financial constraints: Moving beyond the kz index. *Review of Financial Studies*, *23*(5), 1909–1940.

Hassett, K. A., & Hubbard, R. G. (2002). Tax policy and business investment. In *Handbook of public economics* (pp. 1293–1343, Vol. 3).

He, Y., Deng, X., & Zhu, Y. (2019). Impact of VAT Refund Policy on Corporate Value: Empirical Evidence from Chinese Listed Companies [In Chinese]. *Fiscal Research*, (5), 104–117.

Jensen, M. C., & Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, *3*(4), 305–360.

Jorgenson, D. W. (1963). Capital theory and investment behavior. *The American Economic Review*, *53*(2), 247–259.

Kaplan, S. N., & Zingales, L. (1997). Do investment-cash flow sensitivities provide useful measures of financing constraints? *The Quarterly Journal of Economics*, *112*(1), 169–215.

Keynes, J. M. (1936). *The general theory of employment, interest and money.* Macmillan.

Klemm, A. (2021). *The design of tax incentives and their evaluation* (tech. rep.). International Monetary Fund.

Liu, C., Li, A., Wang, S., et al. (2023). The Impact of VAT Refund Policy on Investment and Employment in Manufacturing Enterprises [In Chinese]. *Journal of Central University of Finance and Economics*, (3), 31–42.

Liu, Y., & Mao, J. (2019). How do tax incentives affect investment and productivity? firm-level evidence from china. *American Economic Journal: Economic Policy*, *11*(3), 261–291.

Liu, Z., Liang, H., & Liu, C. (2023). The effects of debt liquidity risk on firms' growth rate. *International Journal of Managerial Finance*, *20*(1), 71–93.

Luo, Y., & Liu, Y. (2021). Corporate Financial Asset Allocation and the Mechanism of Easing Financing Constraints: Based on Liquidity Management Analysis [In Chinese]. *Financial Development Research*, (3), 3–13.

Ma, C. (2010). Construction and Empirical Analysis of Corporate Financial Flexibility Index [In Chinese]. *Systems Engineering*, *28*(10), 61–66.

Ma, H. (2024). Protection of SME Rights and Innovation Activities [In Chinese]. *The World Economy*, *47*(12), 193–218.

Meng, W., Zhong, K., & Guo, X. (2018). Monetary Policy, Liquidity Management and Enterprise Value Creation [In Chinese]. *Finance and Trade Economics*, (8), 55–64.

Miller, D., & Friesen, P. H. (1984). A longitudinal study of the corporate life cycle. *Management Science*, *30*(10), 1161–1183.

Miller, M. H., & Orr, D. (1966). A model of the demand for money by firms. *The Quarterly Journal of Economics*, *80*(3), 413–435.

Modigliani, F., & Miller, M. H. (1958). The cost of capital, corporation finance and the theory of investment. *The American Economic Review*, *48*(3), 261–297.

Myers, S. C., & Majluf, N. S. (1984). Corporate financing and investment decisions when firms have information that investors do not have. *Journal of Financial Economics*, *13*(2), 187–221.

Ni, X., Huang, K., & Peng, Y. (2024). Liquidity Value Support and Corporate Financing Demand: On the Significance of Adequate Liquidity [In Chinese]. *Financial Review*, (2), 34–49.

Ohrn, E. (2019). The effect of tax incentives on u.s. manufacturing: Evidence from state accelerated depreciation policies. *Journal of Public Economics*, *180*, 104123.

Opler, T., Pinkowitz, L., Stulz, R., & Williamson, R. (1999). The determinants and implications of corporate cash holdings. *Journal of Financial Economics*, *52*(1), 3–46.

Rauh, J. D. (2006). Investment and financing constraints: Evidence from the funding of corporate pension plans. *Journal of Finance*, *61*(1), 33–71.

Ross, S. A. (1977). The determination of financial structure: The incentive signalling approach. *The Bell Journal of Economics*, *8*(1), 23–40.

Ross, S. A., Westerfield, R. W., & Jordan, B. D. (2019). *Fundamentals of corporate finance*. McGraw-Hill Education.

Spence, M. (1973). Job market signaling. *Quarterly Journal of Economics*, *87*(3), 355–374.

Tait, A. A. (1988). *Value-added tax: International practice and problems*. International Monetary Fund.

Tax incentives for business investment: A primer for policy makers in developing countries. (2002). *World Development*, *30*(9), 1497–1516.

Tian, L., & Lu, X. (2021). Tax Reduction, Business Entry and Exit, and Total Factor Productivity [In Chinese]. *Management World*, *37*(12), 56–77.

Tian, Z., Jin, S., & Wu, Q. (2024). Impact of VAT Refunds on VAT Revenue: A Discussion on the Automatic Stabilizer Mechanism [In Chinese]. *Fiscal Research*, (8), 66–84.

Wang, M., & Li, M. (2024). VAT Refund Policy, Corporate Liquidity, and Labor Income Enhancement [In Chinese]. *Management World*, *40*(4), 60–88.

Wilson, J. D. (1999). Theories of tax competition. *National Tax Journal*, *52*(2), 269–304.

World Bank. (2020). *Global economic prospects*.

Wu, Y., Lv, C., & Ni, C. (2021). VAT Neutrality, Corporate Investment, and Value: Evidence from the VAT Refund Reform [In Chinese]. *Management World*, *37*(8), 180–194.

Xie, Y., Qin, J., Jin, Z., et al. (2022). VAT Neutrality and Corporate Cash Holdings [In Chinese]. *Trade Economics*, *43*(12), 80–96.

Yagan, D. (2015). Capital tax reform and the real economy: The effects of the 2003 dividend tax cut. *American Economic Review*, *105*(12), 3531–3563.

Yue, S., & Xiao, C. (2023). Can VAT Refunds Alleviate Financing Constraints? Evidence Based on Cash Flow Sensitivity [In Chinese]. *Trade Economics*, *44*(1), 51–67.

Zhang, J. (2020). Monetary Policy, Financing Constraints, and Corporate Liquidity [In Chinese]. *Finance and Economy*, (7), 10–23.

Zhang, Q., & Huang, S. (2019). Can Enterprise Implicit Soft Power Improve Liquidity Constraints? [In Chinese]. *Management World*, *35*(11), 144–157.

Zhang, T., & Liu, W. (2024). VAT Refund Reform, Financing Constraints, and Corporate Supply Chain Linkages [In Chinese]. *Management World*, *40*(3), 94–115.

Zhang, Y., Song, H., & Huang, Y. (2024). VAT Refund and High-Quality Development of Enterprises: Empirical Evidence from Chinese Listed Companies [In Chinese]. *Taxation Research*, (4), 108–116.

Zhao, R., Zhou, X., & Zhang, W. (2024). Tax Incentives, Diversified Operations and Tax Avoidance of Group Companies [In Chinese]. *Journal of Quantitative and Technical Economics*, (8), 93–112.

Zidar, O., Gale, W., & Toder, E. (2018). *Tax reform and its macroeconomic impact* (tech. rep.). NBER.

Zwick, E., & Mahon, J. (2017). Tax policy and heterogeneous investment behavior. *American Economic Review*, *107*(1), 217–248.

A Calculation Procedure of First-Level Indicators of Financial Flexibility

According to C. Ma (2010), in the calculation process of financial flexibility, the indicator system is divided into 3 first-level indicators and 6 second-level indicators. Among them, the weights of the first-level indicators are calculated using the Analytic Hierarchy Process (AHP) (Li (2003)). It should be noted that this process is independent of the raw data itself. The calculation steps are as follows:

A.1 Constructing the Judgment Matrix

When calculating the weights of the first-level indicators, we first need to construct a judgment matrix. This matrix is based on pairwise comparisons of the indicators, and the relative importance of the first-level indicators is evaluated to form the judgment matrix.

For each pair of indicators, we adopt the 1–9 scale method:

- 1: two indicators are equally important,
- 3: one indicator is slightly more important,
- 5: one indicator is important,
- 7: one indicator is strongly important,
- 9: one indicator is extremely important,
- 2, 4, 6, 8: intermediate values between the above levels.

For the three first-level indicators—"Current Indicator", "Leverage Indicator", and "External Financing Cost Indicator"—the judgment matrix is as follows:

$$A = \begin{bmatrix} 1 & \frac{1}{3} & 7 \\ 3 & 1 & 9 \\ \frac{1}{7} & \frac{1}{9} & 1 \end{bmatrix}$$

A.2 Calculating Eigenvalues and Eigenvectors

To obtain the weights of the first-level indicators, we calculate the eigenvalues and eigenvectors of the judgment matrix. The eigenvalue represents the "degree of importance" of the judgment matrix, while the eigenvector represents the weights of each indicator. The calculation steps are as follows:

1. Compute the eigenvalues and eigenvectors of the judgment matrix;
2. Select the eigenvector corresponding to the largest eigenvalue;
3. Normalize the eigenvector (so that the sum equals 1), obtaining the final weights.

A.3 Consistency Test

The consistency test is an important step in the AHP method, used to determine whether the constructed judgment matrix is reasonable. If the judgment matrix is inconsistent, it will affect the accuracy of the final weights. Therefore, a consistency test must be conducted. The Consistency Index (CI) can be calculated as:

$$CI = \frac{\lambda_{\max} - n}{n - 1}$$

where $\lambda_{\max}$ is the largest eigenvalue of the judgment matrix, and n is the order of the matrix.

$$CR = \frac{CI}{RI} \tag{A.1}$$

Here, RI denotes the Random Consistency Index, which can be obtained from standard tables according to the matrix dimension. If the Consistency Ratio (CR) is less than 0.1, the judgment matrix is considered to have good consistency; if $CR \geq 0.1$, it indicates substantial inconsistency, and the judgment matrix needs to be reconstructed.

The calculation yields the weights of the three first-level indicators as 0.65, 0.29, and 0.06, corresponding to the liability side, the cash side, and the financing cost side, respectively. The constructed matrix passes the consistency test with $CI = 0.04 < 0.1$.

B Alternative Fixed Effects and Clustering Strategies

Following the approach of Wang and Li (2024) this paper replaces the baseline model's in section 4 (Equation (2) in the main text) fixed effects and clustering strategies. Specifically, the clustering approach is switched from industry-level clustering—which accounts for cross-industry heterogeneity—to a more granular strategy: while retaining individual and time fixed effects, we replace province-time fixed effects with city-time fixed effects to test the robustness of our results. The revised specification is presented in Equation B.1.1, where C_i denotes city fixed effects:

$$FF_{it} = \beta_0 + \beta_1 DID_{it} + \beta_2 X_{it} + \theta_t + c_i + \gamma_t + \epsilon_{it} \tag{B.1.1}$$

Regression results are reported in Table B.1. Columns (1)–(4) show that the coefficients remain statistically significant, indicating that the DID estimates are robust to the change in fixed effects and clustering at the industry level. In particular, Column (4) controls for additional covariates, and the coefficient on the DID term is 0.0332, statistically significant at the 5% level, suggesting robustness of the results. In terms of the robustness check, the parallel trend test is shown in Figure B.1. Further, we restrict the sample to a matched sample using propensity score matching (PSM-DID, as shown in Table, employing both nearest-neighbor and kernel matching techniques. The results consistently support the findings in Equation (B.1.1), reinforcing the robustness of our conclusions. The placebo test (B.2) shows the same.

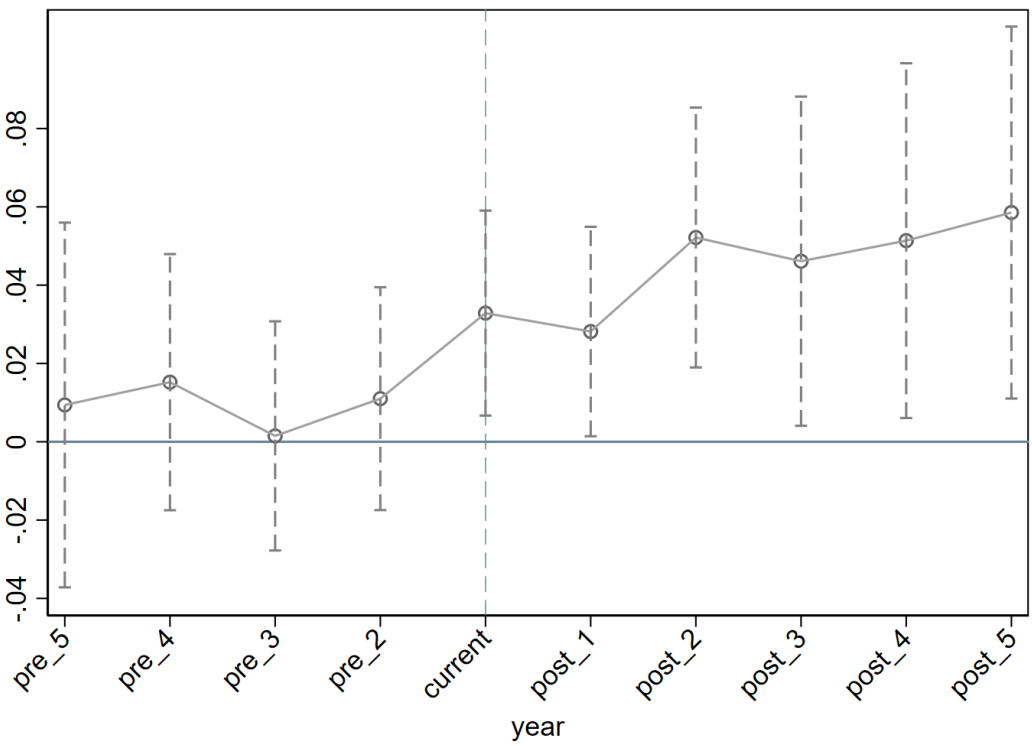

Figure B.1: PSM-DID Nearest-Neighbor Matching Results

Table B.1: Alternative Fixed Effects and Clustering Method

	(1) FF	(2) FF	(3) FF	(4) FF
DID	0.1965***	0.0339**	0.0513***	0.0332**
	(0.0138)	(0.0190)	(0.0109)	(0.0152)
Boardsize			-0.0305	-0.0501
			(0.0224)	(0.0385)
Size			3.6744***	3.6128***
			(0.0398)	(0.1763)
Indp			0.0708	0.0157
			(0.0677)	(0.0661)
Dual			-0.0147**	-0.0153
			(0.0070)	(0.0103)
Salerg			-0.0347**	-0.0205**
			(0.0069)	(0.0091)
ROA			0.0181	0.0100
			(0.0527)	(0.1327)
Tobinq			0.0065***	0.0032
			(0.0024)	(0.0051)
Degree			0.0180	
			(0.0122)	
Constant	10.2189***	10.3101***	1.8613***	2.1500***
	(0.0122)	(0.0049)	(0.1036)	(0.4364)
Individual FE		✓		✓
Time FE		✓		✓
Province*Time FE		✓		✓
No. of Obs	11624	10468	9570	8547
Within-R^2	0.0828	0.8423	0.4090	0.8802

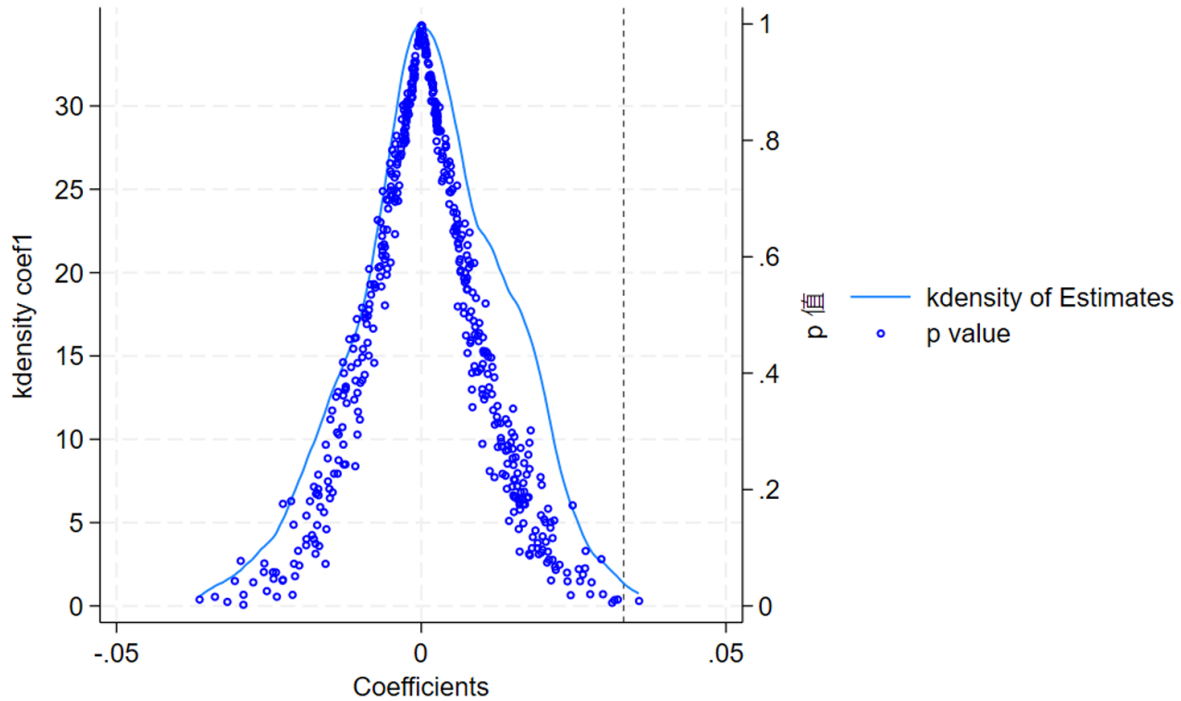

Figure B.2: Placebo Test

Table B.2: PSM-DID Test Result

	NN Matching		Kernel Matching	
	(1)	(2)	(3)	(4)
	FF	FF	FF	FF
DID	0.0190	0.0323**	0.0194	0.0328**
	(0.0183)	(0.0150)	(0.0185)	(0.0152)
Boardsize	0.0301	-0.0448	0.0354	-0.0463
	(0.0385)	(0.0422)	(0.0392)	(0.0391)
Size	3.3998***	3.6403***	3.4403***	3.6181***
	(0.0737)	(0.1751)	(0.0717)	(0.1743)
Indp	0.1166	0.0505	0.1145	0.0263
	(0.0953)	(0.0663)	(0.0968)0	(0.0668)
Dual	-0.0160	-0.0135	-0.0170	-0.0158
	(0.0132)	(0.0104)	(0.0129)	(0.0102)
Salerg	-0.0518**	-0.0203**	-0.0551**	-0.0203**
	(0.00131)	(0.0093)	(0.0130)	(0.0089)
ROA	0.2923*	0.0362	0.2798*	0.0096
	(0.1533)	(0.1331)	(0.1546)	(0.1326)
Tobinq	-0.0002	0.0011	0.0005	0.0021
	(0.0061)	(0.0061)	(0.0062)	(0.0051)
Degree	0.0236**		0.0230**	
	(0.0102)		(0.0104)	
Constant	2.3337***	2.0632***	2.2344***	2.1283***
	(0.0716)	(0.4291)	(0.1812)	(0.4326)
Individual FE		✓		✓
Time FE		✓		✓
Province*Time FE		✓		✓
No. of Obs	8850	7839	9565	8533
Within-R^2	0.7244	0.8665	0.7482	0.8804

C Other Estimator Robustness Checks

C.1 CSDID Estimator

To ensure the robustness of the estimation results, this paper first employs the DID estimation method proposed by Callaway and Sant'Anna (2021), known as the CSDID estimator, to analyze policy effects under staggered adoption and multiple periods. Compared to the conventional TWFE+DID model, the CSDID estimator offers the following advantages: (1) Heterogeneous treatment timing: It allows units to receive treatment at different points in time, making it suitable for evaluating policies implemented in a rolling fashion; (2) More robust estimation of dynamic effects: By aggregating group-time average treatment effects through an appropriate weighting scheme, it provides a more accurate overall estimate of the ATT (Average Treatment effect on the Treated). The core idea is to construct a proper counterfactual path using not-yet-treated groups at each time point and estimate the treatment effects for each group g (i.e., the cohort first treated at time t=g) in period t and subsequent periods. The estimation formula is shown in Equation B.2.1.

$$ATT(g,t) = \mathbb{E}[Y_t(1) - Y_t(0)|G = g] \tag{B.2.1}$$

where $Y_t(1)$ denotes the outcome under treatment and $Y_t(0)$ represents the counterfactual outcome in the absence of treatment. The notation $G = g$ indicates that a unit first receives treatment at time g. The overall ATT is then computed as in equation B.2.2), where the weight w_g is proportional to the sample size of units in group g.

$$ATT_{overall} = \sum_g w_g \cdot (\frac{1}{T_g} \sum_{t \geq g} ATT(g,t)) \tag{B.2.2}$$

This paper adopts the CSDID estimator for policy effect estimation. The estimation results, along with a comparison to the traditional two-way fixed effects (TWFE) DID model, are presented in Table C.1. Columns (1) and (2) replicate the results reported in Section 4.4 of the main text. Column (3) reports the estimated Average Treatment Effect of the Treated Group (ATT) using the CSDID estimator, which is 0.0248 and statistically significant at the 10% level. In contrast, Column (4) estimates the Average Treatment Effect of the Untreated Group (ATU), yielding a coefficient that is not statistically different from zero, indicating that the treated group indeed experienced a significant policy effect.

C.2 Double Machine Learning (DML) Estimator

To verify the robustness of the causal effect of the VAT credit refund policy on firms' financial flexibility, we adopt the Double Machine Learning (DML) approach proposed by Chernozhukov et al. (2018). The DML method is suited for high-dimensional settings, using orthogonalization and cross-fitting to address the curse of dimensionality and model misspecification. First, we consider the following partially linear regression (PLR) model as equation B.2.3 and B.2.4.

$$Y_{it} = D_{it} \cdot \theta_0 + g(X_{it}) + U_{it} \tag{B.2.3}$$

$$D_{it} = m(X_{it}) + V_{it} \tag{B.2.4}$$

where Y_{it} is firm Financial Flexibility i in year t, D_{it} is the treatment indicator for exposure to the VAT credit refund policy. X_{it} is a high-dimensional covariate vector, and $g(X_{it})$ denotes an Unknown non-parametric function. Parameter θ_0 is the Treatment effect we hope to estimate. Finally, the U_{it} and

V_{it} are error terms, satisfying $\mathbb{E}[U_{it}|D_{it}, X_{it}] = 0$ and $\mathbb{E}[V_{it}|D_{it}] = 0$.

We begin by estimating the nuisance functions $\hat{g}(X_{it})$ and $\hat{m}(X_{it})$ and using machine learning methods such as random forests or deep neural networks. Then, we construct a GMM estimator B.2.5.

$$\mathbb{E}[D_{it}(Y_{it} - g(X_{it}) - D\theta_0)] \tag{B.2.5}$$

and it satisfies the condition below.

$$\mathbb{E}[D(Y_{it} - g(X_{it} - D\theta_0))] = \frac{1}{n}\sum_{i=1}^{n} D_{it}(Y_{it} - \hat{g}(X_{it} - D\hat{\theta}) \tag{B.2.6}$$

Next, we construct the Neyman Orthogonalization score (equation B.2.7).

$$\psi(W_{it}; \theta_0; \eta) = (D_{it} - \hat{m}(X_{it})) \cdot (Y_{it} - \hat{g}(X_{it}) - \theta_0 \cdot (D_{it} - \hat{m}(X_{it}))) \tag{B.2.7}$$

where $W_{it} = [Y_{it}, X_{it}, D_{it}]$, $\eta = [g(X_{it}), m(X_{it})]$, and $(D_{it} - \hat{m}(X_{it}))$, and $(Y_{it} - \hat{g}(X_{it}) - \theta_0 \cdot (D_{it} - \hat{m}(X_{it})))$ are the estimation of the error terms V and U respectively. Based on that, we plug the Neyman orthogonalization score in to the GMM estimator, and obtain:

$$\hat{\theta} = [\frac{1}{n}\sum_{i=1}^{n} D_{it} - \hat{m}(X_{it})]^{-1}[\frac{1}{n}\sum_{i=1}^{n}(D_{it} - \hat{m}(X_{it})(Y_{it} - \hat{g}(X_{it})))] = [\frac{1}{n}\sum_{i=1}^{n}\hat{V}_{it}^2]^{-1}(\frac{1}{n}\sum_{i=1}^{n}\hat{V}_{it}\hat{U}_{it}) \tag{B.2.8}$$

According to the Central Limit Theorem (CLT) and the Weak Law of Large Number (WLLN), the estimator satisfies both the consistency and the Asymptotic Normality in large sample:

$$\sqrt{n}(\widehat{\theta}_{DML} - \theta_0) \to \mathcal{N}(0, \sigma^2) \tag{B.2.9}$$

where $\sigma^2 = A^{-1}\mathbb{E}[\psi\psi^T]A^{-1^T}$, and $A = \frac{1}{n}\sum \partial\theta\varphi(Y_{it}, X_{it}, D_{it}; g_0(X), m_0(X); \theta_0)$.

Since the DML model requires cross-sectional data, this paper transforms the panel dataset accordingly. Specifically, for each firm, the change in firm Financial Flexibility is calculated as the difference between the post-policy average and the pre-policy average. Control variables are taken from the year prior to the policy implementation (year 2017 in this case), and the treatment indicator (Treat) is retained. As reported in Column (5) of Table C.1, the estimated coefficient is 0.0273, which is positive and statistically significant at a 1% level. This result suggests that the causal effect of the policy remains robust, further reinforcing the credibility of the paper's main conclusions.

Table C.1: Test Result

	(1) TWFE+DID	(2) TWFE+DID	(3) CSDID(ATT)	(4) CSDID(ATT)	(5) DML
DID	0.0317**	0.0263**	0.0248*	0.0063	0.0273***
	(0.0134)	(0.0130)	(0.0145)	(0.0058)	(0.0080)
Controls		✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	
Time FE	✓	✓	✓	✓	
Province*Time FE	✓	✓	✓	✓	
No. of Obs	11624	9570	9213	9213	
Within-R^2	0.2387	0.4254			

D Heterogeneity Tests: Extended Analysis

D.1 Basic Principles

Causal Forest is a machine learning method designed to estimate Heterogeneous Treatment Effects (HTE). It originates from the Generalized Random Forest (GRF) framework proposed by Wager and Athey (2018). Building on traditional Random Forest regression trees, it embeds the logic of causal inference to capture the heterogeneity in the response of the outcome variable to the "treatment" variable. In other words, it addresses the core question: "How does an intervention affect different individuals differently?" The central task is to estimate the Conditional Average Treatment Effect (CATE), that is, the marginal effect of an intervention variable W on an outcome variable Y, conditional on a particular feature vector $X = x$ (Athey and Imbens (2016)). Formally, CATE is expressed as:

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]. \tag{4}$$

Here, $Y(1)$ and $Y(0)$ denote the potential outcomes for the treatment and control groups, respectively. Causal Forests achieve unbiased estimation of treatment effect heterogeneity by employing honesty splitting within each regression tree. Specifically, the training sample is divided into two independent subsamples: one used for determining the split points, and the other used for estimating treatment effects. This procedure effectively reduces overfitting and ensures unbiased heterogeneity estimates.

Compared with traditional linear interaction models or subgroup regressions, causal forests possess unique advantages in handling high-dimensional covariates and complex nonlinear relationships. First, they do not require the manual specification of variables; instead, the data automatically identifies and ranks the most significant sources of heterogeneity. Second, causal forests are robust to multicollinearity among covariates and maintain good estimation accuracy even in noisy environments. Moreover, they allow for transparent decomposition of each tree's contribution to the estimated treatment effects, thereby enabling more intuitive and controllable variable importance analysis.

Most importantly, the method supports subgroup analysis by partitioning the sample according to key variables into two or more groups. This enables researchers to uncover conclusions such as "policy effects are stronger among highly leveraged firms" or "no significant effects are found for firms with lower ROA." Such insights are particularly valuable for policy evaluation and design. By avoiding model specification errors that commonly arise in traditional regression models when dealing with complex structural heterogeneity, causal forests provide a flexible and robust, data-driven solution.

D.2 Empirical Analysis

This section applies the causal forest method to evaluate the impact of the VAT credit refund policy on firms' financial flexibility. In the study of heterogeneity, the selection of covariates is particularly critical, as they must operate within the causal mechanism "VAT credit refund policy → improvement in firms' financial flexibility." In other words, covariates should plausibly constitute links in the causal chain. Furthermore, their meanings should not be overly similar (for example, firm size and fixed asset investment share a certain degree of similarity).

Accordingly, in terms of variables, this paper defines firms' eligibility for the 2018 and 2022 policy implementations as the treatment variable T; the outcome variable is firms' financial flexibility (FF). For covariates, the control variables presented in Section 4.1 are subjected to a round of replacement and screening, resulting in the final selection of the following eight variables:

1. Firm Size (Size)

2. Return on Assets (ROA)
3. Leverage Ratio (LEV)
4. Executives' Education Level (Degree)
5. Tobin Q
6. SA Index
7. Z-Score
8. Board Size

To ensure robustness in model training and evaluation, the data are randomly split into an 80% training set and a 20% test set. The training set is used to construct the causal forest model, while the test set is reserved for prediction and validation. In model construction, following Sylvia et al. (2020), we employ the grf package in R to import the outcome variable, control variables, and the treatment dummy. We set the number of trees to 1,000 to enhance estimation stability and adopt *honest estimation*, dividing the sample into training and estimation subsets to mitigate overfitting risk.

After training, the accuracy and robustness of the model are evaluated on the test set to verify its ability to effectively capture heterogeneity in treatment effects. Table D.1 presents the ATE estimates from both traditional multi-period DID models and the causal forest. Columns (3) and (4) incorporate control variables and fixed effects: Column (3) includes province-by-time fixed effects with clustering at the firm level, while Column (4) includes city-by-time fixed effects with clustering at the industry level. In both specifications, the coefficients remain significantly positive. Column (5) reports the causal forest results, where the estimated coefficient is significantly positive at the 5% level, corroborating the robustness of both the regression results and the article's main findings.

The next step involves ranking the variable importance of each covariate to identify which play a stronger role in the causal mechanism "VAT credit refund policy → improved financial flexibility." The ranking results are presented in Table D.2. Firm size emerges as the most important variable; the Z-Score follows closely, suggesting that firms at the margin of financial distress are more sensitive to the refund policy. Leverage ratio ranks next, indicating heterogeneous uses of refund funds depending on capital structure: highly leveraged firms may prioritize debt repayment, thereby rapidly improving financial flexibility. Tobin's Q and executives' education rank lower, reflecting that under widespread financial stress, market expectations and managerial quality exert weakened effects. High-valuation firms may also prefer debt repayment over expansion, while the education effect could exhibit diminishing marginal returns given the relative homogeneity of managerial teams; accordingly, board size ranks last. In addition, ROA and cash flow receive low rankings since they are closer to the outcome variable itself rather than serving as mechanisms or moderators, and thus exhibit weaker causal effects.

Table D.1: Multi-period DID and Factor Forest Regression Results

	(1) FF	(2) FF	(3) FF	(4) FF	(5) FF
DID	-0.0222*	-0.0222	0.0219**	0.0246*	
	(0.0117)	(0.0205)	(0.0111)	(0.0128)	
CF					0.0276**
					(0.0112)
Boardsize	0.0299	0.0299	-0.0317	-0.0372	
	(0.0288)	(0.0336)	(0.0279)	(0.0345)	
Size	3.3912***	3.3912***	3.4785***	3.4427***	
	(0.0643)	(0.0888)	(0.1440)	(0.2256)	
SA	0.0173	0.0173	-0.1109	-0.1547	
	(0.0235)	(0.0258)	(0.0853)	(0.1021)	
LEV	11.1963	11.1963	-28.4076***	-28.8428**	
	(7.8072)	(8.9450)	(10.2614)	(13.4429)	
ROA	0.2052**	0.2052	0.0032	0.0183	
	(0.0872)	(0.1403)	(0.0789)	(0.1303)	
ZScore	-0.0027	-0.0027	-0.0055***	-0.0061***	
	(0.0017)	(0.0017)	(0.0015)	(0.0012)	
Tobinq	0.0086	0.0086	0.0193***	0.0214***	
	(0.0066)	(0.0090)	(0.0055)	(0.0056)	
Degree	0.0474***	0.0474***			
	(0.0154)	(0.0117)			
Controls	✓	✓	✓	✓	✓
Firm FE			✓	✓	
Time FE			✓	✓	
Province*Time FE			✓		
City*Time FE				✓	
Clustered by	Firm	Industry	Firm	Industry	None
No. of Obs	9900	9900	9809	8855	–
Within-R^2	0.7554	0.7554	0.8810	0.8826	–

Table D.2: Variable Importance Ranking

Rank	Variable	Importance
1	Size	23.231%
2	TobinQ	20.265%
3	Degree	18.731%
4	Z-Score	12.721%
5	LEV	9.892%
6	SA	6.714%
7	ROA	6.120%
8	Boardsize	2.325%

Figure D.1 illustrates how the CATE varies with different values of each covariate, along with the fitted curves. A clear negative relationship is observed between firm size and CATE, with a pronounced threshold around Size = 2.25. Beyond this threshold, the improvement in financial flexibility diminishes, suggesting that smaller firms benefit more significantly from the VAT credit refund policy. By contrast, CATE exhibits a strong positive relationship with both the Z-Score and Tobin's Q, indicating that financially distressed firms and growth-oriented firms experience larger improvements in financial flexibility. Leverage ratio also shows a positive association with CATE, implying that firms with higher debt ratios realize greater improvements in flexibility. For other variables, such as executives' education, ROA, and board size, CATE remains relatively unchanged across different values, suggesting limited moderating

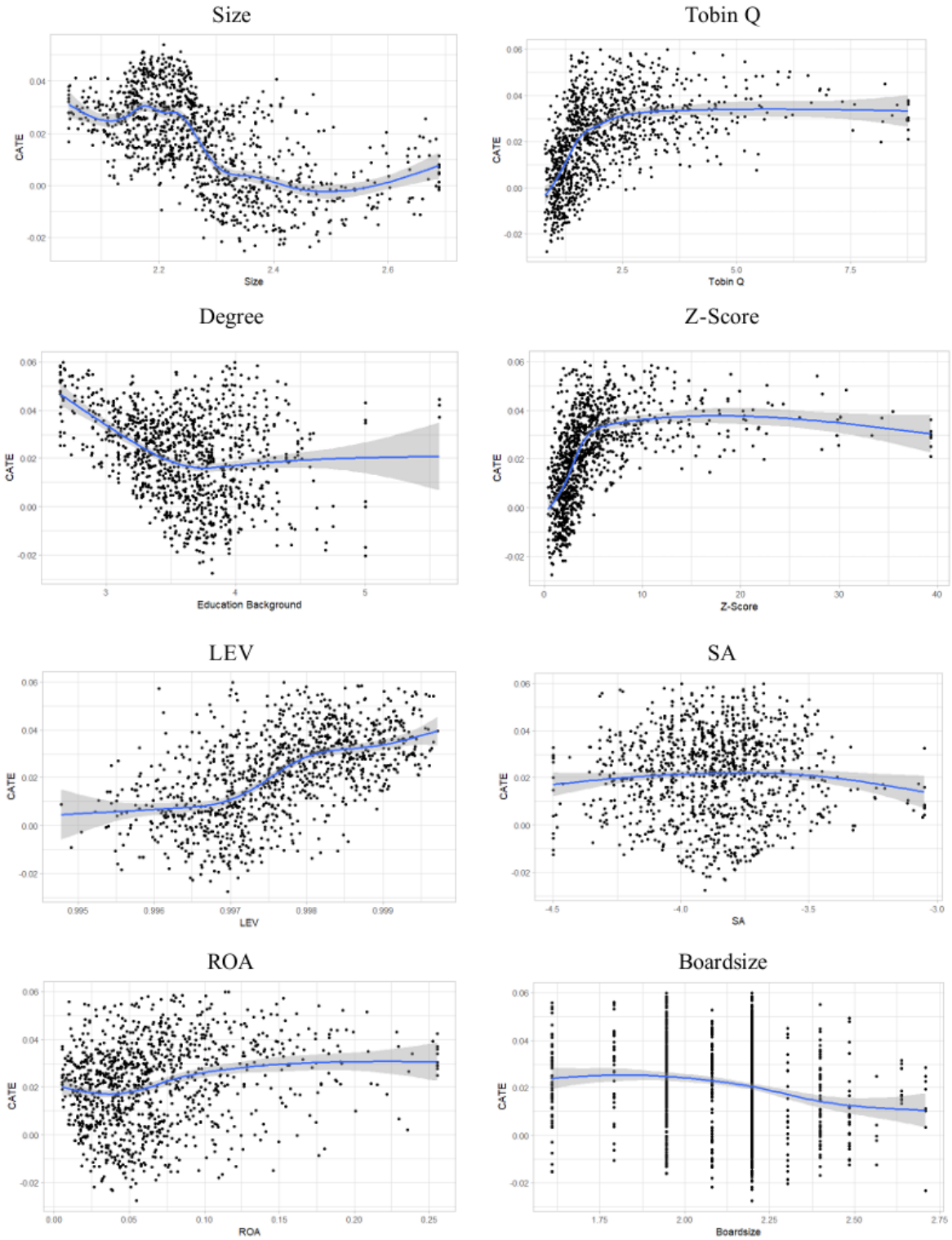

Figure D.1: CATE Estimation on all Covariate

effects. Figure C.2 presents the structure of a representative causal tree and the sample partitioning strategy.

To verify the statistical significance of heterogeneity estimated by the causal forest, we follow the heterogeneity testing procedure described in Section 6.3 of the main text. We select the top five covariates from Table C.2—Size, Tobin Q, Degree, Z-Score, and LEV—for validation. For Size and LEV, we create binary variables ("VariableName_High") based on whether their values exceed the median, and interact them with the policy variable (DID). For Tobin Q and Z-Score, we create indicator variables for the top 10% quantile, while for Degree we follow the same approach as in Section 6.3. The heterogeneity identification model is constructed in line with the specification in Section 6.3.

The regression results are reported in Tables D.3 and D.4. Table D.3 augments the two-way fixed effects specification with province-by-time fixed effects and clusters standard errors at the firm level, while Table D.4 replaces these with city-by-time fixed effects and clusters at the industry level. Both tables show consistent findings: smaller firms experience larger improvements in financial flexibility; firms with higher leverage ratios also benefit more significantly. Similarly, financially less stable firms display stronger gains. By contrast, heterogeneity related to executives' education and firm growth potential is not statistically significant.

Table D.3: Heterogeneity Analysis – 1

	(1)	(2)	(3)	(4)	(5)
	FF	FF	FF	FF	FF
Size_High × DID	-0.0459**				
	(0.0181)				
Degree_High × DID		-0.0044			
		(0.0257)			
LEV_High × DID			0.0297*		
			(0.0154)		
Tobinq_High × DID				-0.0126	
				(0.0266)	
ZScore_High × DID					0.0737**
					(0.0303)
DID	0.0479***	0.0273**	0.0098	0.0273**	0.0218*
	(0.0146)	(0.0122)	(0.0136)	(0.0114)	(0.0116)
Controls	Yes	Yes	Yes	Yes	Yes
FE (Ind. & Time)	Yes	Yes	Yes	Yes	Yes
Province × Time FE	Yes	Yes	Yes	Yes	Yes
N	9900	9900	9900	9900	9900
Adj-R^2	0.4264	0.4252	0.4269	0.4253	0.4264

Table D.4: Heterogeneity Analysis – 2

	(1) FF	(2) FF	(3) FF	(4) FF	(5) FF
Size_High*DID	-0.0536**				
	(0.0204)				
Degree_High*DID		-0.0032			
		(0.0262)			
LEV_High*DID			0.0390***		
			(0.0127)		
Tobinq_High*DID				-0.0030	
				(0.0299)	
ZScore_High*DID					0.0674***
					(0.0226)
DID	0.0585***	0.0338*	0.0118	0.0329**	0.0283*
	(0.0191)	(0.0179)	(0.0168)	(0.0144)	(0.0150)
N	8547	8547	8547	8547	8547
Adj-R^2	0.8805	0.8801	0.8807	0.8802	0.8803

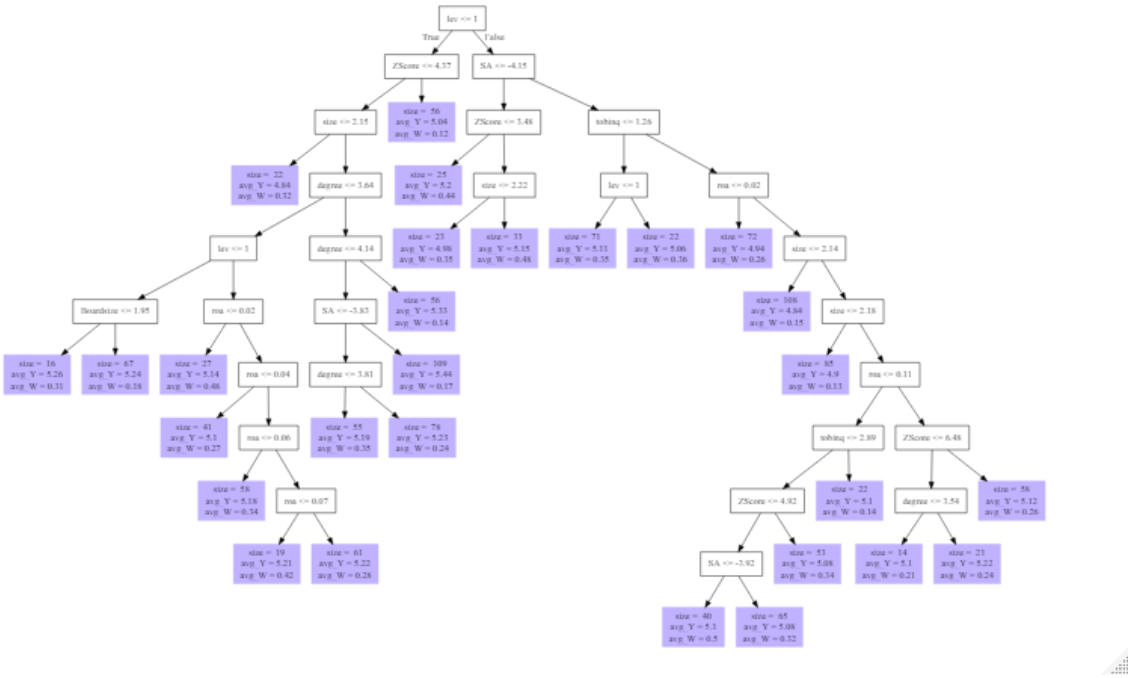

Figure D.2: Sample Tree

References for Appendix A-D

Athey, S., & Imbens, G. W. (2016). Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, *113*(27), 7353–7360.

Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, *225*(2), 200–230.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, *21*(1), C1–C68.

Li, W. (2003). Design and Application of China's Corporate Governance Index [In Chinese]. *Proceedings of the Third Annual Conference of the Chinese Economists Society*, 6–9.

Sylvia, S., Warrinnier, N., Luo, R., Yue, A., Attanasio, O., Medina, A., & Rozelle, S. (2020). From quantity to quality: Delivering a home-based parenting intervention through china's family planning cadres. *The Economic Journal*, *131* (April), 1365–1400.

E Announcement on Scope of Industries Eligible for VAT Credit Refund

Ministry of Finance & State Taxation Administration, P.R.China

Announcement No. 70, 2018

Article 1: Scope of Industries Eligible for Refund

Industries eligible for VAT credit refunds include advanced manufacturing (such as equipment manufacturing), modern service industries (such as R&D), and power grid enterprises, as specified below:

1. **Advanced manufacturing and modern service industries.** According to the industrial classification of the national economy, these include 18 broad industry categories such as special equipment manufacturing and R&D, as listed in the attached document *"2018 Catalogue of Industries Eligible for VAT Credit Refunds."* Taxpayers' industries are determined by their registered classification. Priority is given to:

   (a) The ten priority areas defined in *Made in China 2025*, such as next-generation information technology, high-end CNC machine tools and robots, aerospace equipment, marine engineering equipment and high-tech vessels, advanced rail transit, energy-saving and new energy vehicles, power equipment, agricultural machinery, new materials, and biomedicine and high-performance medical devices.

   (b) High-tech enterprises, technologically advanced service enterprises, and technology-based SMEs.

2. **Power grid enterprises.** All enterprises holding electricity operation licenses (transmission or distribution).

Article 2: Conditions for Taxpayers

Taxpayers must have a VAT credit rating of Grade A or B.

Article 3: Calculation of Refundable VAT Credits

Taxpayers apply to their competent tax authority. Refundable VAT credits for the current period are determined by the ending balance of the previous period and the refund ratio, subject to the cap of the taxpayer's VAT credit balance as of end-2017:

1. Refundable VAT credits = Ending balance of VAT credits in the previous period × Refund ratio.

2. Refund ratio is calculated as:

   (a) For taxpayers registered on or before December 31, 2014: the ratio of deductible input VAT (from invoices, import VAT payment certificates, etc.) in 2015–2017 to total deductible input VAT in the same period.

   (b) For taxpayers registered on or after January 1, 2015: the ratio of deductible input VAT during the actual operation period to total deductible input VAT in the same period.

3. If the refundable VAT credits do not exceed the end-2017 balance, the refund equals the refundable amount. If they exceed the 2017 balance, the refund is capped at the 2017 balance.

Article 4: Work Requirements

1. Provincial finance and tax departments shall determine eligible taxpayers within their jurisdiction based on the refund quotas set by MOF and STA, balancing enterprise scale and type. Lists of taxpayers and refund amounts must be reported to MOF and STA for record by August 31, 2018. Power grid enterprise refunds are calculated directly under this Notice.

2. Provincial authorities shall carefully plan, coordinate, monitor, and publicize the refund process, ensuring smooth and orderly implementation, and complete refunds by September 30, 2018.

3. By October 31, 2018, provincial authorities shall submit summary reports on refund implementation, including results, methods, effects, and suggestions. Major issues shall be promptly reported to MOF and STA.

Ministry of Finance, State Taxation Administration
June 27, 2018

Attachment: 2018 Catalogue of Industries Eligible for VAT Credit Refunds

No.	Industry (Correspond to the National Economic Industry Classification)
1	Manufacture of chemical raw materials and chemical products
2	Manufacture of pharmaceuticals
3	Manufacture of chemical fibers
4	Manufacture of non-metallic mineral products
5	Manufacture of metal products
6	Manufacture of general equipment
7	Manufacture of special equipment
8	Manufacture of automobiles
9	Manufacture of railway, ship, aerospace, and other transport equipment
10	Manufacture of electrical machinery and apparatus
11	Manufacture of computers, communications, and other electronic equipment
12	Manufacture of instruments and meters
13	Internet and related services
14	Software and information technology services
15	Research and experimental development
16	Professional technical services
17	Science and technology promotion and application services
18	Ecological protection and environmental governance

F Announcement on Further Enhancing the Implementation of the VAT Credit Refund Policy at Period-End

Ministry of Finance & State Taxation Administration, P.R.China

Announcement No. 14, 2022

To support the development of small and micro enterprises and industries such as manufacturing, boost market confidence, and stimulate vitality among market entities, the following policies on further strengthening the VAT credit refund at period-end are announced[12]:

Article 1: VAT Credit Refunds for Small and Micro Enterprises

The policy of full monthly refund of incremental VAT credits for advanced manufacturing is expanded to eligible small and micro enterprises (including sole proprietorships). In addition, a one-off refund of stock VAT credits will be granted.

1. Eligible small and micro enterprises may apply for incremental VAT credit refunds starting from the April 2022 tax filing period. Refund conditions shall be implemented in accordance with Article 3 of this Announcement before December 31, 2022.

2. Eligible micro enterprises may apply for a one-off refund of stock VAT credits beginning April 2022; eligible small enterprises may apply starting May 2022.

Article 2: VAT Credit Refunds for Manufacturing and Related Industries

The scope of monthly full refunds is expanded to enterprises in manufacturing, scientific research and technical services, utilities (electricity, heat, gas, water), software and information technology services, ecological protection and environmental governance, and transportation, warehousing, and postal services. These enterprises are also eligible for a one-off refund of stock credits.

1. Eligible enterprises in these industries may apply for incremental VAT credit refunds starting from April 2022.

2. Eligible medium-sized enterprises may apply for a one-off refund of stock VAT credits from July 2022; eligible large enterprises may apply from October 2022.

Article 3: Eligibility Conditions

Taxpayers applying for refunds under this Announcement must meet all of the following:

1. VAT credit rating of Grade A or B;

2. No fraudulent VAT credit refunds, export tax refunds, or issuance of false VAT invoices within 36 months prior to application;

3. No more than one penalty for tax evasion within 36 months prior to application;

4. Not having enjoyed "levy and refund upon collection" policies since April 1, 2019.

[12] For Original Document, see: https://www.gov.cn/zhengce/zhengceku/2022-03/22/content_5680600.htm.

Article 4: Definition of Incremental VAT Credits

1. Before a one-off refund of stock VAT credits, incremental VAT credits equal the period-end VAT credit balance minus the balance as of March 31, 2019.
2. After a one-off refund, incremental VAT credits equal the period-end VAT credit balance.

Article 5: Definition of Stock VAT Credits

1. Before a one-off refund, stock VAT credits equal the March 31, 2019 balance if the period-end balance is greater; otherwise equal to the current period-end balance.
2. After a one-off refund, stock VAT credits equal zero.

Article 6–15: Additional Provisions

- Classification of enterprises (micro, small, medium, large) based on revenue and assets, following MIIT [2011] No. 300 and PBOC [2015] No. 309.
- Definition of eligible industries: VAT sales share from listed industries must exceed 50%.
- Calculation of refundable credits:

$$\text{Refundable Incremental Credit} = \text{Incremental Credit} \times \text{Input Ratio} \times 100\%$$

$$\text{Refundable Stock Credit} = \text{Stock Credit} \times \text{Input Ratio} \times 100\%$$

- Interaction with export refund policies, prohibition on duplicate benefits, and conditions for repayment.
- Administrative arrangements: refunds to be processed in batches by April 30, June 30, September 30, and December 31, 2022.

Article 16: Effective Date

This Announcement shall take effect on April 1, 2022. Previous announcements (MOF & STA 2019 No. 84; 2020 No. 2, Article 6; and 2021 No. 15) are repealed simultaneously.

Ministry of Finance & State Taxation Administration
March 21, 2022